

periodic rotating motion, which is represented mathematically by a sinusoid. Second, all periodic waveforms (e.g., sawtooth, triangle, square) can be reconstructed as the sum of component sinusoidal waves (Fourier's theorem).

Sinusoidal signals (voltages and currents) have three basic characteristics: amplitude (or magnitude), frequency, and phase (or phase angle). The frequency of all voltages and currents in an AC circuit is constant, uniform, and determined by an independent voltage or current source. As a result, it is not necessary to compute frequency in phasor analysis. On the other hand, while the amplitude and phase of voltages and currents in an AC circuit are constant, they are not uniform and are determined not only by an independent source but also by the elements present in the circuit. Consequently, AC circuit analysis is concerned with the computation of the amplitude and phase of one or more voltages and currents. (DC circuit analysis was only concerned with amplitude.) Phasor analysis is well-suited for AC circuit analysis because a phasor bundles together amplitude and phase into a single quantity known as a phasor.

In phasor analysis, resistors, capacitors, and inductors are represented as impedance elements. Impedance allows Ohm's law to be generalized as a phasor relationship applicable to resistors, capacitors, and inductors. Kirchhoff's laws can also be generalized as phasor relationships. Consequently, AC circuits can be solved using the same DC methods (e.g., voltage division, current division, nodal analysis, superposition, Thévenin's and Norton's theorem, and source transformations) discussed in Chapters 2 and 3. The only difference is that these relationships now involve phasors, that is, complex quantities.

The average and effective (root-mean-square) amplitude of a waveform are introduced in this chapter. An effective value represents the equivalent DC value required to supply or dissipate the same power as the AC waveform and thus provides a means of comparing different waveforms.

In this chapter and throughout the book, angles are given in units of radians, unless indicated otherwise.

## Learning Objectives

1. Compute current, voltage, and energy of capacitors and inductors. *Section 4.1.*
2. Calculate the average and effective (root-mean-square) value of an arbitrary periodic waveform. *Section 4.2.*
3. Write the differential equation(s) for circuits containing inductors and capacitors. *Section 4.3.*
4. Convert time-domain sinusoidal voltages and currents to phasor notation, and vice versa; and represent circuits using impedances. *Section 4.4.*
5. Apply DC circuit analysis methods to AC circuits in phasor form. *Section 4.5.*

## 4.1 CAPACITORS AND INDUCTORS

The ideal resistor was introduced in Chapter 2 as a useful approximation of many practical electrical devices. However, in addition to resistance, which always dissipates energy, an electric circuit may also exhibit capacitance and inductance, which act to store and release energy, in the same way that an expansion tank and flywheel,

respectively, act in a mechanical system. These two distinct energy storage mechanisms are represented in electric circuits by two ideal circuit elements: the ideal capacitor and the ideal inductor, which approximate the behavior of actual discrete capacitors and inductors. They also approximate the bulk properties of capacitance and inductance that are present in any physical system. In practice, any element of an electric circuit will exhibit some resistance, some inductance, and some capacitance, that is, some ability to dissipate and store energy.

The energy of a capacitor is stored within the electric field between two conducting plates while the energy of an inductor is stored within the magnetic field of a conducting coil. Both elements can be charged (i.e., stored energy is increased) or discharged (i.e., stored energy is decreased). Ideal capacitors and inductors can store energy indefinitely; however, in practice, discrete capacitors and inductors exhibit "leakage," which typically results in a gradual reduction in the stored energy over time.

All the relationships for capacitors and inductors exhibit duality, which means that the capacitor relations are mirror images of the inductor relations. Examples of duality are apparent in Table 4.1.

Table 4.1 Properties of capacitors and inductors

|                        | Capacitors                                              | Inductors                                               |
|------------------------|---------------------------------------------------------|---------------------------------------------------------|
| Differential $i$ - $v$ | $i = C \frac{dv}{dt}$                                   | $v = L \frac{di}{dt}$                                   |
| Integral $i$ - $v$     | $v_C(t) = \frac{1}{C} \int_{-\infty}^t i_C(\tau) d\tau$ | $i_L(t) = \frac{1}{L} \int_{-\infty}^t v_L(\tau) d\tau$ |
| DC equivalent          | Open-circuit                                            | Short-circuit                                           |
| Two in series          | $C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$                    | $L_{eq} = L_1 + L_2$                                    |
| Two in parallel        | $C_{eq} = C_1 + C_2$                                    | $L_{eq} = \frac{L_1 L_2}{L_1 + L_2}$                    |
| Stored energy          | $W_C = \frac{1}{2} C v_C^2$                             | $W_L = \frac{1}{2} L i_L^2$                             |

## The Ideal Capacitor

A capacitor is a device that can store energy due to a charge separation. In general, a capacitor (and thus, capacitance) is present when any two conducting surfaces are separated by a distance. A simple example is two parallel plates of shared cross-sectional area  $A$  separated by a distance  $d$ . The gap between the plates may be a vacuum or filled with some dielectric material, such as air, mica, or Teflon. The impact of the dielectric material on the capacitance is represented by the dielectric constant  $\kappa$ .<sup>1</sup> Figure 4.2 depicts a typical configuration and the circuit symbol for a capacitor.

The capacitance  $C$  of an ideal parallel-plate capacitor such as the one described above is:

$$C = \frac{\kappa \epsilon_0 A}{d} \quad (4.1)$$

where  $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$  is the permittivity constant of a vacuum.

<sup>1</sup> A dielectric material is a material that is not an electrical conductor but contains a large number of electric dipoles, which become polarized in the presence of an electric field.



## Hydraulic Analog of a Capacitor

If the walls of a vessel have some elasticity, energy is stored in the walls when the vessel is filled by a fluid or gas (e.g., an inflated balloon). The ratio of the mass of the fluid or gas to the potential energy stored in the walls per unit mass is the fluid capacitance of the vessel, a property similar to electrical capacitance. Figure 4.1 depicts a gas bag accumulator, such as an expansion tank attached to the hot water line in many residential homes. The two-chamber arrangement permits fluid to displace a membrane separating an incompressible fluid (e.g., water) from a compressible fluid (e.g., air). The analogy shown in Figure 4.1 assumes that the reference pressure  $p_0$  and the reference voltage  $v_2$  are both zero.

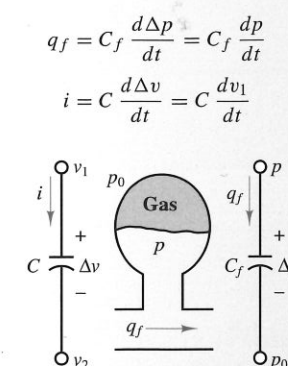
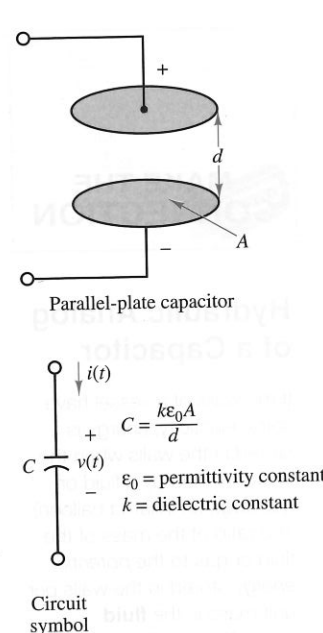


Figure 4.1 Analogy between electrical and fluid capacitance



**Figure 4.2** Structure of parallel-plate capacitor

The presence of a dielectric or vacuum between the conducting plates does not permit charge to pass directly from one plate to the other. However, if the applied voltage across a capacitor changes so will the accumulated charge. Thus, although no charge can literally pass from one plate of an ideal capacitor directly through to the other, a change in voltage will cause the accumulated charge to change, which is the equivalent effect of a current through the capacitor.

$$Q_C = CV_C \quad (4.2)$$

where the parameter  $C$  is the capacitance and is a measure of the ability of the device to accumulate charge. The unit of capacitance is coulomb per volt, or farad (F). The farad is an impractically large unit for many common electronic applications; units of microfarads ( $1 \mu\text{F} = 10^{-6} \text{ F}$ ) and picofarads ( $1 \text{ pF} = 10^{-12} \text{ F}$ ) are more common in practice.

The current through a capacitor is defined as the time rate of change of its stored charge. That is,

$$i_C(t) = \frac{dq_C(t)}{dt} \quad (4.3)$$

The  $i$ - $v$  relationship for a capacitor is obtained from equation 4.3 by using equation 4.2 to plug in for  $q_C(t)$ . The result is:

$$i_C(t) = C \frac{dv_C(t)}{dt} \quad i\text{-}v \text{ relation for capacitor} \quad (4.4)$$

Equation 4.4 can be integrated to yield an equivalent  $i$ - $v$  relationship for a capacitor:

$$v_C(t) = \frac{1}{C} \int_{-\infty}^t i_C(\tau) d\tau \quad (4.5)$$

One immediate implication of equation 4.4 is that the current through a capacitor in a DC circuit is zero. Why? Since the voltage across a capacitor in a DC circuit must, by definition, be constant, the time derivative of the voltage must be zero. Thus, equation 4.4 requires the current through the capacitor to also be zero.

A capacitor in a DC circuit is equivalent to an open-circuit.

Equation 4.5 indicates that the voltage across a capacitor depends on the history of the current through it. To calculate that voltage it is necessary to know the initial voltage  $V_0$  (i.e., an initial condition) across the capacitor at some previous time  $t_0$ . Then:

$$v_C(t) = V_0 + \frac{1}{C} \int_{t_0}^t i_C(\tau) d\tau \quad t \geq t_0 \quad (4.6)$$

The significance of the initial voltage  $v_C(t_0) = V_0$  is simply that at time  $t_0$  some charge was stored in the capacitor, resulting in  $V_0$ , according to the relationship  $Q = CV$ .

### Equivalent Capacitance

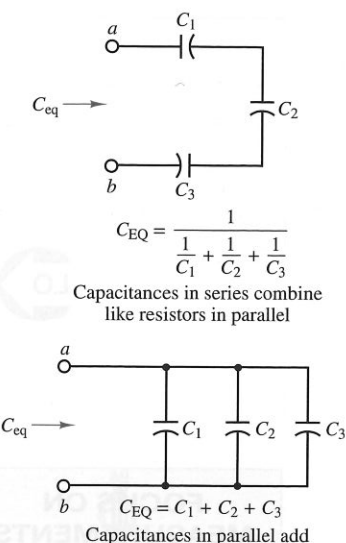
Just as resistors can be in series and parallel to yield an equivalent resistance, so capacitors can also be in series and parallel to yield an equivalent capacitance. The calculation rules are given below.

For two capacitors in series and parallel, the equivalent capacitances are, respectively:

$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2} \quad \text{and} \quad C_{eq} = C_1 + C_2 \quad (4.7)$$

Notice that the rule for the equivalent capacitance of two capacitors in series is the product divided by the sum, which is the same rule used for two resistors in parallel. Likewise, the equivalent capacitance of two capacitors in parallel is simply the sum of the two, which is the same rule used for two resistors in series. The more general rules are illustrated in Figure 4.3.

When calculating equivalent capacitance, capacitors in series combine like resistors in parallel and capacitors in parallel combine like resistors in series.



**Figure 4.3** Equivalent capacitance in a circuit

### Discrete Capacitors

Actual capacitors are rarely constructed of two parallel plates separated by air because this configuration either yields very low values of capacitance or requires very large plate areas. To increase the capacitance (i.e., the ability to store energy), physical capacitors are often made of tightly rolled sheets of metal film, with a dielectric (e.g., paper or Mylar) sandwiched in between. Table 4.2 illustrates typical values, materials, maximum voltage ratings, and useful frequency ranges for various types of capacitors. The voltage rating is important because any insulator will break down if a sufficiently high voltage is applied across it.

**Table 4.2** Capacitors

| Material     | Capacitance range                       | Maximum voltage (V) | Frequency range (Hz) |
|--------------|-----------------------------------------|---------------------|----------------------|
| Mica         | 1 pF to 0.1 $\mu\text{F}$               | 100–600             | $10^3$ – $10^{10}$   |
| Ceramic      | 10 pF to 1 $\mu\text{F}$                | 50–1,000            | $10^3$ – $10^{10}$   |
| Mylar        | 0.001 $\mu\text{F}$ to 10 $\mu\text{F}$ | 50–500              | $10^2$ – $10^8$      |
| Paper        | 1,000 pF to 50 $\mu\text{F}$            | 100–10,000          | $10^2$ – $10^8$      |
| Electrolytic | 0.1 $\mu\text{F}$ to 0.2 F              | 3–600               | $10$ – $10^4$        |

In practice, actual capacitors exhibit some leakage between the plates. Imperfect construction techniques invariably provide some capability for charge to pass from one plate to the other. This imperfection is often represented by an equivalent resistance in parallel with an ideal capacitor.

### Energy Storage in Capacitors

The energy stored in a capacitor  $W_C(t)$  may be derived easily from its definition as the time integral of power, which is the product of voltage and current:

$$P_C(t) = i_C(t)v_C(t) = C \frac{dv_C(t)}{dt} v_C(t) = \frac{d}{dt} \left[ \frac{1}{2} C v_C^2(t) \right] \quad (4.8)$$

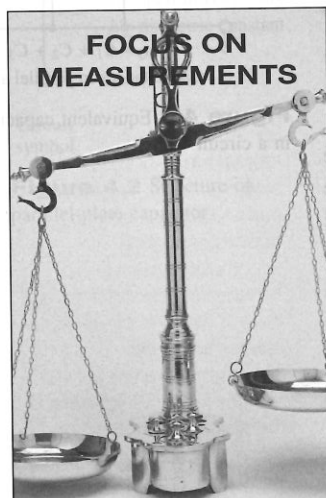
The total energy stored in the inductor is found by integrating the power, as shown below:

$$W_C(t) = \int P_C(\tau) d\tau = \int \frac{d}{d\tau} \left[ \frac{1}{2} C v_C^2(\tau) \right] d\tau \quad (4.9)$$



$$W_C(t) = \frac{1}{2} C v_C^2(t) \quad \text{Energy, in joules, stored in a capacitor}$$

Example 4.4 illustrates the calculation of the energy stored in a capacitor.



### Capacitive Displacement Transducer and Microphone

As shown in Figure 4.2, the capacitance of a flat parallel-plate capacitor is:

$$C = \frac{\epsilon A}{d} = \frac{\kappa \epsilon_0 A}{d}$$

where  $\epsilon$  is the **permittivity** of the dielectric material,  $\kappa$  is the dielectric constant,  $\epsilon_0 = 8.854 \times 10^{-12}$  F/m is the permittivity of a vacuum,  $A$  is the area of each of the plates, and  $d$  is their separation. The dielectric constant for air is  $\kappa_{\text{air}} \approx 1$ . Thus, the capacitance of two flat parallel plates of area  $1 \text{ m}^2$ , separated by a 1-mm air gap is 8.854 nF, a very small value for such large plates. As a result, flat parallel-plate capacitors are impractical for use in most electronic devices. On the other hand, parallel-plate capacitors find application as motion transducers, that is, as devices that can measure the motion or displacement of an object. In a capacitive motion transducer, the plates are designed to allow relative motion when subjected to an external force. Using the capacitance value just derived for a parallel-plate capacitor, one can obtain the expression

$$C = \frac{8.854 \times 10^{-3} A}{x}$$

where  $C$  is the capacitance in picofarads,  $A$  is the area of the plates in square millimeters, and  $x$  is the separation distance in millimeters. Note that the change in  $C$  due to a change in  $x$  is non-linear, since  $C \propto 1/x$ . However, for small changes in  $x$ , the change in  $C$  is approximately linear.

The sensitivity  $S$  of the transducer is defined as the rate of change in capacitance  $C$  with respect to a change in separation distance  $x$ .

$$S = \frac{dC}{dx} = -\frac{8.854 \times 10^{-3} A}{x^2} \quad \frac{\text{pF}}{\text{mm}}$$

Thus, the sensitivity is itself a function of the separation distance, as shown in Figure 4.4. Note that as  $x \rightarrow 0$ , the slope of  $C(x)$  increases and so the sensitivity  $S$  increases as well. Figure 4.4 depicts this behavior for a transducer with area equal to  $10 \text{ mm}^2$ . This type of capacitive displacement transducer is used in the popular condenser microphone, in which sound pressure waves act to displace a thin metallic foil. The change in capacitance can then be converted to a change in voltage or current by means of a suitable circuit. An extension of this concept that permits measurement of differential pressures is shown in Figure 4.5. A three-terminal variable capacitor is made of two fixed surfaces with a

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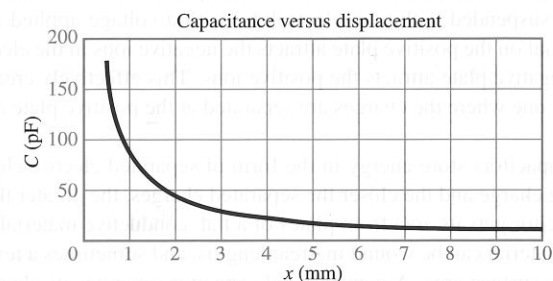


Figure 4.4 Response of a capacitive displacement transducer

single deflecting plate, often made of steel, between them. Typically, the fixed surfaces are spherical depressions ground into glass disks and coated with a conducting material. Inlet orifices expose the deflecting plate to the outside fluid or gas. When the pressure on both sides of the deflecting plate is the same, the capacitance between terminals  $b$  and  $d$ , denoted by  $C_{db}$ , will be equal to that between terminals  $b$  and  $c$ , denoted by  $C_{bc}$ . If any pressure differential exists, the two capacitances will change, with an increase on the side where the deflecting plate has come closer to the fixed surface and a corresponding decrease on the other side.

A Wheatstone bridge circuit, such as that shown in Figure 4.5, is ideally suited to precisely balance the output voltage  $v_{\text{out}}$  when the differential pressure across the transducer is zero. The output voltage will deviate from zero whenever the two capacitances are not equal because of a pressure differential across the transducer. The bridge circuit will be analyzed later.

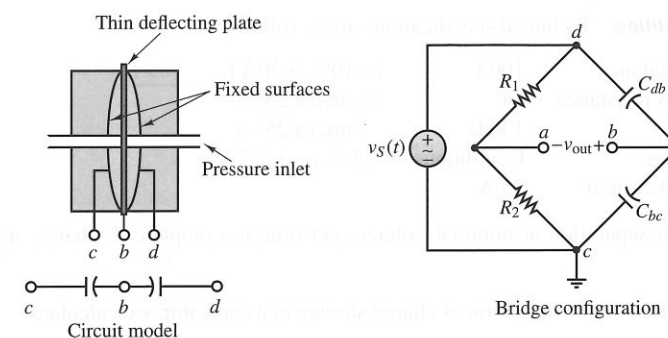


Figure 4.5 Capacitive pressure transducer and related bridge circuit

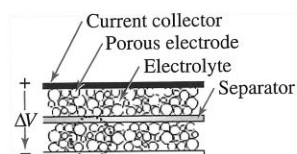
### EXAMPLE 4.1 Charge Separation in Ultracapacitors

#### Problem

Ultracapacitors are finding application in a variety of fields, including as a replacement or supplement for batteries in hybrid-electric vehicles. In this example you will make your first acquaintance with these devices.

An ultracapacitor, or “supercapacitor,” stores energy electrostatically by polarizing an electrolytic solution. Although it is an electrochemical device (also known as an electrochemical





**Figure 4.6** Ultracapacitor structure

double-layer capacitor), there are no chemical reactions involved in its energy storage mechanism. This mechanism is highly reversible, allowing the ultracapacitor to be charged and discharged hundreds of thousands of times. An ultracapacitor can be viewed as two nonreactive porous plates suspended within an electrolyte, with a voltage applied across the plates. The applied potential on the positive plate attracts the negative ions in the electrolyte while the potential on the negative plate attracts the positive ions. This effectively creates two layers of capacitive storage, one where the charges are separated at the positive plate and another at the negative plate.

Recall that capacitors store energy in the form of separated electric charge. The greater the area for storing charge and the closer the separated charges, the greater the capacitance. A conventional capacitor gets its area from plates of a flat, conductive material. To achieve high capacitance, this material can be wound in great lengths, and sometimes a texture is imprinted on it to increase its surface area. A conventional capacitor separates its charged plates with a dielectric material, sometimes a plastic or paper film, or a ceramic. These dielectrics can be made only as thin as the available films or applied materials.

An ultracapacitor gets its area from a porous carbon-based electrode material, as shown in Figure 4.6. The porous structure of this material allows its surface area to approach 2,000 square meters per gram ( $\text{m}^2/\text{g}$ ), much greater than can be accomplished using flat or textured films and plates. An ultracapacitor's charge separation distance is determined by the size of the ions in the electrolyte, which are attracted to the charged electrode. This charge separation [less than 10 angstroms ( $\text{\AA}$ )] is much smaller than can be achieved using conventional dielectric materials. The combination of enormous surface area and extremely small charge separation gives the ultracapacitor its outstanding capacitance relative to conventional capacitors.

Use the data provided to calculate the charge stored in an ultracapacitor and calculate how long it will take to discharge the capacitor at the maximum current rate.

#### Solution

**Known Quantities:** Technical specifications are as follows:

|                   |            |                   |
|-------------------|------------|-------------------|
| Capacitance       | 100 F      | (−10%/+30%)       |
| Series resistance | DC         | 15 mΩ(±25%)       |
|                   | 1 kHz      | 7 mΩ(±25%)        |
| Voltage           | Continuous | 2.5 V; peak 2.7 V |
| Rated current     | 25 A       |                   |

**Find:** Charge separation at nominal voltage and time to complete discharge at maximum current rate.

**Analysis:** Based on the definition of charge storage in a capacitor, we calculate

$$Q = CV = 100 \text{ F} \times 2.5 \text{ V} = 250 \text{ C}$$

To calculate how long it would take to discharge the ultracapacitor, approximate the current as:

$$i = \frac{dq}{dt} \approx \frac{\Delta q}{\Delta t}$$

Since the available charge is 250 C, the time to completely discharge the capacitor, assuming a constant 25-A discharge, is:

$$\Delta t = \frac{\Delta q}{i} = \frac{250 \text{ C}}{25 \text{ A}} = 10 \text{ s}$$

**Comments:** Ultracapacitors will be explored further in Chapter 5. The charging and discharging behavior of these devices is examined, taking into consideration their internal resistance.

### EXAMPLE 4.2 Calculating Capacitor Current from Voltage

#### Problem

Calculate the current through a capacitor from knowledge of its terminal voltage.

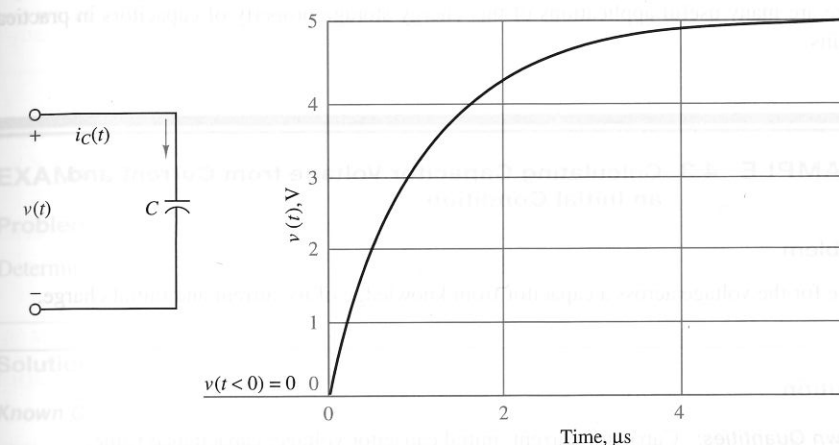
#### Solution

**Known Quantities:** Capacitor terminal voltage; capacitance value.

**Find:** Capacitor current.

**Assumptions:** The initial current through the capacitor is zero.

**Schematics, Diagrams, Circuits, and Given Data:**  $v(t) = 5(1 - e^{-t/10^{-6}}) \text{ V}$ ;  $t \geq 0 \text{ s}$ ;  $C = 0.1 \mu\text{F}$ . The terminal voltage is plotted in Figure 4.7.



**Figure 4.7**

**Assumptions:** The capacitor is initially discharged:  $v(t = 0) = 0$ .

**Analysis:** Using the defining differential relationship for the capacitor, we may obtain the current by differentiating the voltage:

$$i_C(t) = C \frac{dv(t)}{dt} = 10^{-7} \frac{5}{10^{-6}} \left( e^{-t/10^{-6}} \right) = 0.5 e^{-t/10^{-6}} \text{ A} \quad t \geq 0$$

A plot of the capacitor current is shown in Figure 4.8. Note how the current jumps to 0.5 A instantaneously as the voltage rises exponentially: The ability of a capacitor's current to change instantaneously is an important property of capacitors.

**Comments:** As the voltage approaches the constant value 5 V, the capacitor reaches its maximum charge storage capability for that voltage (since  $Q = CV$ ) and no more current flows through the capacitor. The total charge stored is  $Q = 0.5 \times 10^{-6} \text{ C}$ . This is a fairly small amount of charge, but it can produce a substantial amount of current for a brief time. For example, the fully charged capacitor could provide 100 mA of current for a time equal to 5  $\mu\text{s}$ :

$$I = \frac{\Delta Q}{\Delta t} = \frac{0.5 \times 10^{-6}}{5 \times 10^{-6}} = 0.1 \text{ A}$$

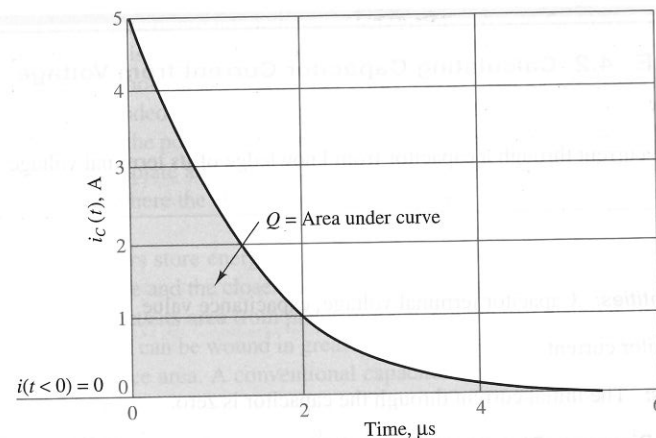


Figure 4.8

There are many useful applications of this energy storage property of capacitors in practical circuits.



### EXAMPLE 4.3 Calculating Capacitor Voltage from Current and an Initial Condition

#### Problem

Solve for the voltage across a capacitor from knowledge of its current and initial charge.

#### Solution

**Known Quantities:** Capacitor current; initial capacitor voltage; capacitance value.

**Find:** Capacitor voltage as a function of time.

**Schematics, Diagrams, Circuits, and Given Data:**

$$i_C(t) = I \begin{cases} 0 & t < 0 \text{ s} \\ 10 \text{ mA} & 0 \leq t \leq 1 \text{ s} \\ 0 & t > 1 \text{ s} \end{cases}$$

$$v_C(t = 0) = 2 \text{ V} \quad C = 1,000 \mu\text{F}$$

The capacitor current is plotted in Figure 4.9(a).

**Assumptions:** The capacitor is initially charged such that  $V_0 = v_C(t = 0) = 2 \text{ V}$ .

**Analysis:** The integral relationship between voltage and current for a capacitor can be used to find voltage when current is known.

$$v_C(t) = \frac{1}{C} \int_{t_0}^t i_C(t') dt' + v_C(t_0) \quad t \geq t_0$$

$$= \begin{cases} \frac{1}{C} \int_0^t I dt' + V_0 = \frac{I}{C} t + V_0 = 10t + 2 \text{ V} & 0 \leq t \leq 1 \text{ s} \\ 12 \text{ V} & t > 1 \text{ s} \end{cases}$$

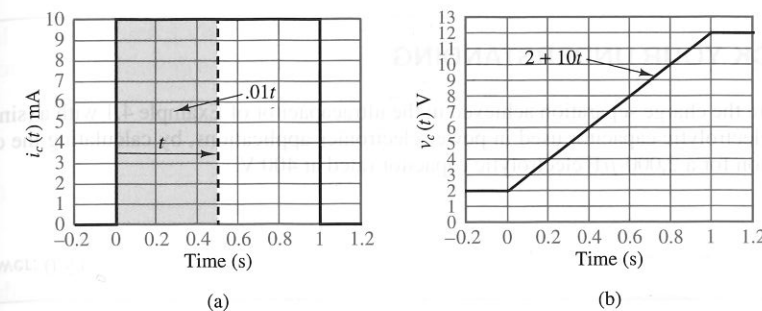


Figure 4.9

**Comments:** Once the current stops, at  $t = 1 \text{ s}$ , the capacitor voltage remains constant because the charge remains constant. That is,  $V = Q/C = \text{constant} = 12 \text{ V}$  at  $t = 1 \text{ s}$ . Remember, the final value of the capacitor voltage depends on two factors: (1) the initial value of the capacitor voltage and (2) the history of the capacitor current. Figure 4.9(a) and (b) depict the two waveforms.

### EXAMPLE 4.4 Energy Storage in Ultracapacitors

#### Problem

Determine the energy stored in the ultracapacitor of Example 4.1.

#### Solution

**Known Quantities:** See Example 4.1.

**Find:** Energy stored in capacitor.

**Analysis:** To calculate the energy, use equation 4.9:

$$W_C = \frac{1}{2} C v_C^2 = \frac{1}{2} (100 \text{ F})(2.5 \text{ V})^2 = 312.5 \text{ J}$$

### CHECK YOUR UNDERSTANDING

Compare the energy stored in the ultracapacitor of Example 4.4 with a (similarly sized) electrolytic capacitor used in power electronics applications. Calculate the energy stored for a  $2,000\text{-}\mu\text{F}$  electrolytic capacitor rated at  $400 \text{ V}$ .

Answer: 160 J

## CHECK YOUR UNDERSTANDING

Compare the charge separation achieved in the ultracapacitor of Example 4.1 with a (similarly sized) electrolytic capacitor used in power electronics applications, by calculating the charge separation for a 2,000- $\mu\text{F}$  electrolytic capacitor rated at 400 V.

Answer: 0.8 C

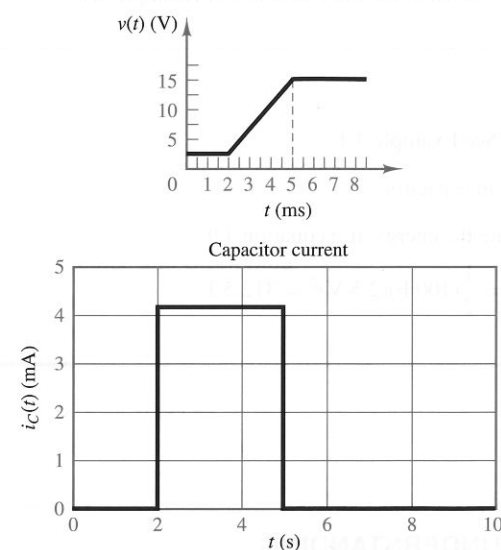
## CHECK YOUR UNDERSTANDING

Find the maximum current through the capacitor of Example 4.3 if the capacitor voltage is described by  $v_C(t) = 5t + 3$  V for  $0 \leq t \leq 5$  s.

Answer: 5 mA

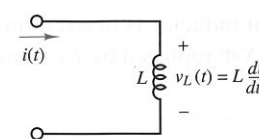
## CHECK YOUR UNDERSTANDING

The voltage waveform shown below appears across a 1,000- $\mu\text{F}$  capacitor. Plot the capacitor current  $i_C(t)$ .

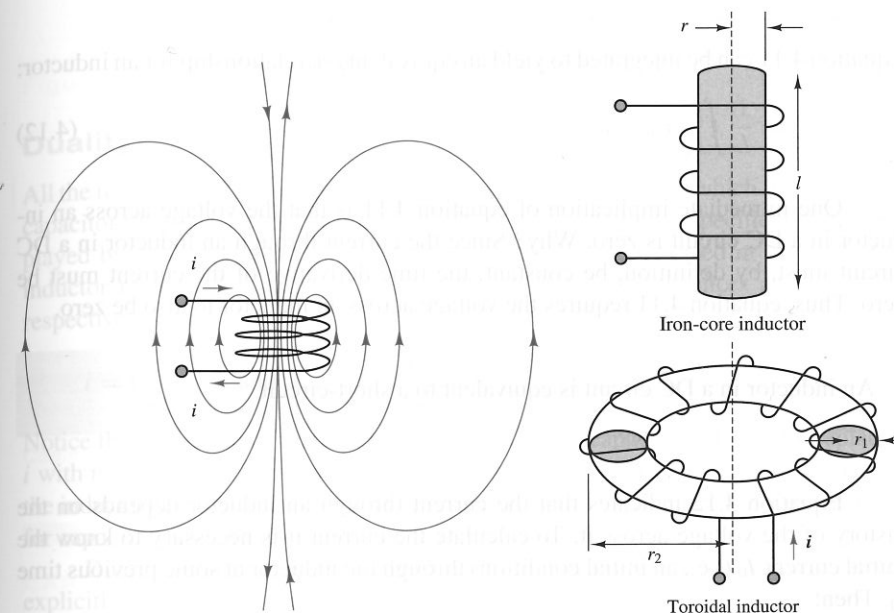


## The Ideal Inductor

An inductor is an element that can store energy in a magnetic field within and around a conducting coil. In general, an inductor (and thus, inductance) is present whenever a conducting wire is turned to form a loop. A simple example is a solenoid, which is a narrow and tightly wound coil of length  $\ell$ , cross-sectional area  $A$ , and  $N$  turns. Inductors are typically made by winding wire around a **core**, which can be an



(a) Circuit symbol



(b) Magnetic flux lines in the vicinity of a current-carrying coil

(c) Practical inductors

Figure 4.10 Inductance and practical inductors

insulator or a ferromagnetic material, as shown in Figure 4.10. A current through the coil establishes a magnetic field through and around the core.<sup>2</sup> In an ideal inductor, the resistance of the wire is zero.

The inductance  $L$  is defined by the following ratio:

$$L \equiv \frac{N\Phi}{i_L}$$

where  $\Phi$  is the magnetic flux through the inductor core and  $i_L$  is the current through the inductor coil. The inductance of an ideal solenoid is:

$$L = \frac{\mu N^2 A}{\ell}$$

where  $\mu$  is the permeability of the core. Another inductor found in many applications is the toroid, which is also depicted in Figure 4.10. Expressions for the inductance of toroids with rectangular and circular cross sections are readily found.

The inductance of a coil is measured in henrys (H) where

$$1 \text{ H} = 1 \text{ V}\cdot\text{s}/\text{A} \quad (4.10)$$

Henrys are reasonable units for practical inductors although millihenrys (mH) are very common and microhenrys ( $\mu\text{H}$ ) are occasionally found.

<sup>2</sup>See Chapter 16.

The  $i$ - $v$  relationship for an inductor is derived directly from Faraday's law of induction but with the total flux  $N\Phi$  replaced by  $Li$  from the definition of inductance  $L$ . The result is:



$$v_L(t) = L \frac{di_L(t)}{dt} \quad i\text{-}v \text{ relation for inductor} \quad (4.11)$$

Equation 4.11 can be integrated to yield an equivalent  $i$ - $v$  relationship for an inductor:

$$i_L(t) = \frac{1}{L} \int_{-\infty}^t v_L(\tau) d\tau \quad (4.12)$$

One immediate implication of equation 4.11 is that the voltage across an inductor in a DC circuit is zero. Why? Since the current through an inductor in a DC circuit must, by definition, be constant, the time derivative of the current must be zero. Thus, equation 4.11 requires the voltage across an inductor to also be zero.

An inductor in a DC circuit is equivalent to a short-circuit.

Equation 4.12 indicates that the current through an inductor depends on the history of the voltage across it. To calculate the current it is necessary to know the initial current  $I_0$  (i.e., an initial condition) through the inductor at some previous time  $t_0$ . Then:

$$i_L(t) = I_0 + \frac{1}{L} \int_{t_0}^t v_L(\tau) d\tau \quad t \geq t_0 \quad (4.13)$$

### Equivalent Inductance

Just as resistors can be in series and parallel to yield an equivalent resistance, so inductors can also be in series and parallel to yield an equivalent inductance. The calculation rules are given below.

For two inductors in series and parallel, the equivalent inductances are, respectively,

$$L_{eq} = L_1 + L_2 \quad \text{and} \quad L_{eq} = \frac{L_1 L_2}{L_1 + L_2} \quad (4.14)$$

Notice that the equivalent inductance of two inductors in series is simply the sum of the two, which is the same rule used for two resistors in series. Likewise, the rule for the equivalent inductance of two inductors in parallel is the product divided by the sum, which is the same rule used for two resistors in parallel. The more general rules are illustrated in Figure 4.11.



When calculating equivalent inductance, inductors in series combine like resistors in series and inductors in parallel combine like resistors in parallel.

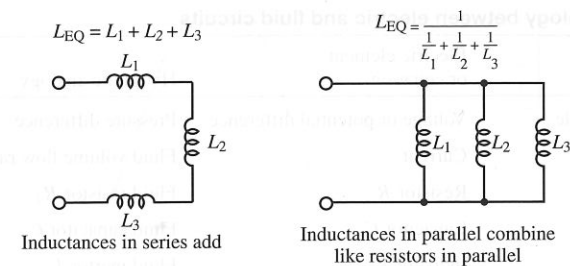


Figure 4.11 Equivalent inductance in a circuit

### Duality

All the relationships for capacitors and inductors exhibit duality, which means that the capacitor relations are mirror images of the inductor relations. Specifically, the roles played by voltage and current in a capacitor relation are reversed in the analogous inductor relation. For example, the  $i$ - $v$  relationships for capacitors and inductors, respectively, are:

$$i = C \frac{dv}{dt} \quad \text{and} \quad v = L \frac{di}{dt}$$

Notice that the inductor relation is obtained from the capacitor relation by replacing  $i$  with  $v$  and  $v$  with  $i$ . It is also necessary, of course, to replace the capacitance  $C$  with the inductance  $L$ . Another example of duality is found in the energy storage relations for capacitors and inductors (see the next section).

Duality is also at work in other relations not involving voltage and current explicitly. For example, consider the rules for calculating equivalent capacitance and equivalent inductance. Capacitors in series combine like inductors in parallel while capacitors in parallel combine like inductors in series. Another example of duality is seen in the DC behavior of capacitors and inductors. In a DC circuit, a capacitor acts like an open-circuit while an inductor acts like a short-circuit. Other examples of duality can be found in later chapters. Duality provides a powerful mnemonic for students trying to memorize equations for a test!

### Energy Storage in Inductors

The energy stored in an inductor  $W_L(t)$  may be derived easily from its definition as the time integral of power, which is the product of voltage and current:

$$P_L(t) = i_L(t)v_L(t) = i_L(t)L \frac{di_L(t)}{dt} = \frac{d}{dt} \left[ \frac{1}{2} Li_L^2(t) \right] \quad (4.15)$$

$$W_L(t) = \int P_L(\tau) d\tau = \int \frac{d}{d\tau} \left[ \frac{1}{2} Li_L^2(\tau) \right] d\tau \quad (4.16)$$

$$W_L(t) = \frac{1}{2} Li_L^2(t) \quad \text{Energy, in joules, stored in an inductor}$$

Note, once again, the duality with the expression for the energy stored in a capacitor, in equation 4.9.



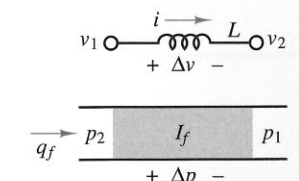
### Hydraulic Analog of an Inductor

**Fluid inertance**, which is caused by the inertial properties (i.e., the mass) of a fluid in motion, is analogous to inductance in an electric circuit. It is well known that a particle in motion possesses kinetic energy; likewise, a fluid in motion, which consists of a collection of particles, also possesses (i.e., stores) kinetic energy. Think of the water flowing out of a fire hose! The equations that define the analogy are given below:

$$\Delta p = p_1 - p_2 = I_f \frac{dq_f}{dt}$$

$$\Delta v = v_1 - v_2 = L \frac{di}{dt}$$

The figure below depicts the analogy. These analogies and the energy equations that apply to electric and fluid circuit elements are summarized in Table 4.3.



Analogy between fluid inertance and electrical inductance

Table 4.3 Analogy between electric and fluid circuits

| Property                 | Electric element or equation    | Hydraulic analogy             |
|--------------------------|---------------------------------|-------------------------------|
| Potential variable       | Voltage or potential difference | Pressure difference           |
| Flow variable            | Current                         | Fluid volume flow rate        |
| Resistance               | Resistor $R$                    | Fluid resistor $R_f$          |
| Capacitance              | Capacitor $C$                   | Fluid capacitor $C_f$         |
| Inductance               | Inductor $L$                    | Fluid inductor $I_f$          |
| Power dissipation        | $P = i^2 R$                     | $P_f = q_f^2 R_f$             |
| Potential energy storage | $W_p = \frac{1}{2} C v^2$       | $W_p = \frac{1}{2} C_f p^2$   |
| Kinetic energy storage   | $W_k = \frac{1}{2} L i^2$       | $W_k = \frac{1}{2} I_f q_f^2$ |

**EXAMPLE 4.5 Calculating Inductor Voltage from Current****Problem**

Calculate the voltage across an inductor from knowledge of its current.

**Solution**

**Known Quantities:** Inductor current; inductance value.

**Find:** Inductor voltage.

**Schematics, Diagrams, Circuits, and Given Data:**

$$i_L(t) = \begin{cases} 0 \text{ mA} & t \leq 1 \text{ ms} \\ -\frac{0.1}{4} + \frac{0.1}{4}t \text{ mA} & 1 \leq t \leq 5 \text{ ms} \\ 0.1 \text{ mA} & 5 \leq t \leq 9 \text{ ms} \\ 13 \times \frac{0.1}{4} - \frac{0.1}{4}t \text{ mA} & 9 \leq t \leq 13 \text{ ms} \\ 0 \text{ mA} & t \geq 13 \text{ ms} \end{cases}$$

$$L = 10 \text{ H}$$

where time  $t$  is in milliseconds. The inductor current is plotted in Figure 4.12.

**Assumptions:**  $i_L(t=0) \leq 0$ .

**Analysis:** The voltage across the inductor is obtained by differentiating the current and multiplying by the inductance  $L$ .

$$v_L(t) = L \frac{di_L(t)}{dt}$$

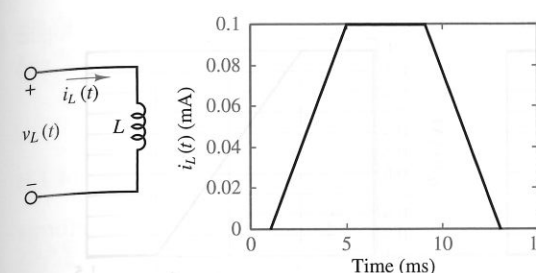


Figure 4.12

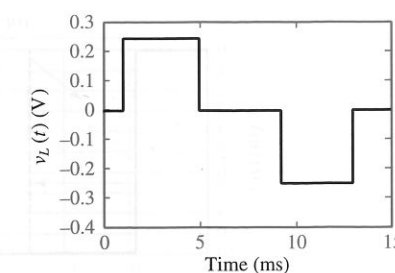


Figure 4.13

Piecewise differentiating the expression for the inductor current, we obtain

$$v_L(t) = \begin{cases} 0 \text{ V} & t < 1 \text{ ms} \\ 0.25 \text{ V} & 1 < t < 5 \text{ ms} \\ 0 \text{ V} & 5 < t < 9 \text{ ms} \\ -0.25 \text{ V} & 9 < t < 13 \text{ ms} \\ 0 \text{ V} & t > 13 \text{ ms} \end{cases}$$

The inductor voltage is plotted in Figure 4.13.

**Comments:** Note how the inductor voltage has the ability to change instantaneously!

**EXAMPLE 4.6 Calculating Inductor Current from Voltage****Problem**

Use a time plot of the voltage across an inductor and its initial current to calculate the current through it as a function of time.

**Solution**

**Known Quantities:** Inductor voltage; initial condition (current at  $t = 0$ ); inductance value.

**Find:** Inductor current.

**Schematics, Diagrams, Circuits, and Given Data:**

$$v(t) = \begin{cases} 0 \text{ V} & t < 0 \text{ s} \\ -10 \text{ mV} & 0 < t < 1 \text{ s} \\ 0 \text{ V} & t > 1 \text{ s} \end{cases}$$

$$L = 10 \text{ mH}; \quad i_L(t=0) = I_0 = 0 \text{ A}$$

The voltage across the inductor is plotted in Figure 4.14(a).

**Analysis:** Use the integral  $i$ - $v$  relationship for an inductor to obtain the current through it:

$$i_L(t) = i_L(t_0) + \frac{1}{L} \int_{t_0}^t v(\tau) d\tau \quad t \geq t_0$$

$$= \begin{cases} I_0 + \frac{1}{L} \int_0^t (-10 \times 10^{-3}) d\tau = 0 + \frac{-10^{-2}}{10^{-2}} t = -t \text{ A} & 0 \leq t \leq 1 \text{ s} \\ -1 \text{ A} & t \geq 1 \text{ s} \end{cases}$$

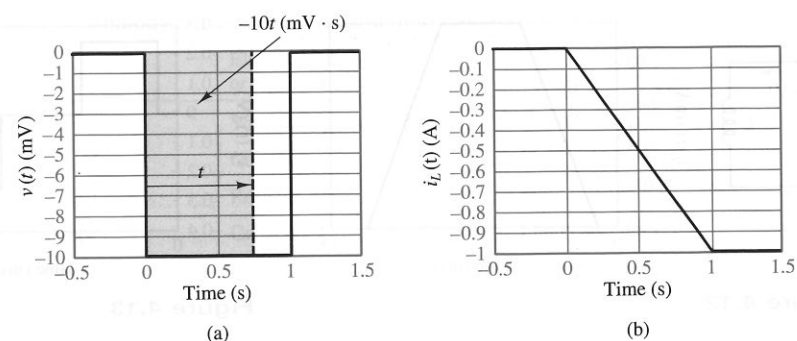


Figure 4.14

The inductor current is plotted in Figure 4.14b.

**Comments:** Note that the inductor voltage can change instantaneously!



### EXAMPLE 4.7 Energy Storage in an Ignition Coil

#### Problem

Determine the energy stored in an automotive ignition coil.

#### Solution

**Known Quantities:** Inductor current initial condition (current at  $t = 0$ ); inductance value.

**Find:** Energy stored in inductor.

**Schematics, Diagrams, Circuits, and Given Data:**  $L = 10 \text{ mH}$ ;  $i_L = I_0 = 8 \text{ A}$ .

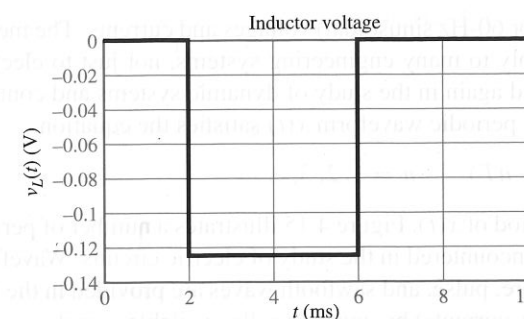
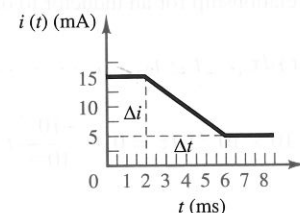
**Analysis:**

$$W_L = \frac{1}{2} L i_L^2 = \frac{1}{2} \times 10^{-2} \times 64 = 32 \times 10^{-2} = 320 \text{ mJ}$$

**Comments:** A more detailed analysis of an automotive ignition coil is presented in Chapter 5 to accompany the discussion of transient voltages and currents.

### CHECK YOUR UNDERSTANDING

The waveform below shows the current through a 50-mH inductor. Plot the inductor voltage  $v_L(t)$ .



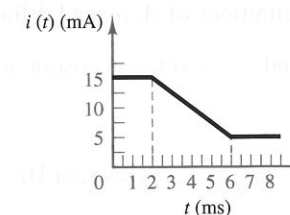
### CHECK YOUR UNDERSTANDING

Find the maximum voltage across a 10-mH inductor when the inductor current is  $i_L(t) = -2t(t - 2) \text{ A}$  for  $0 \leq t \leq 2 \text{ s}$  and zero otherwise.

Answer: 40 mV

### CHECK YOUR UNDERSTANDING

Calculate and plot the inductor energy and power for a 50-mH inductor subject to the current waveform shown below. What is the energy stored at  $t = 3 \text{ ms}$ ?



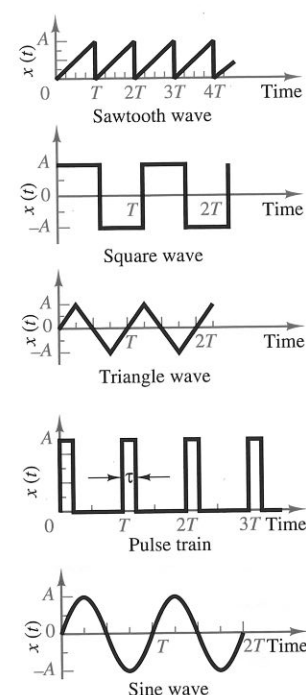
$$p(t) = \begin{cases} 0 & t < 0 \\ (20 \times 10^{-3} - 2.5t)(-0.125 \text{ W}) & 0 \leq t \leq 6 \text{ ms} \\ 0 & t > 6 \text{ ms} \end{cases}$$

$$w(t) = \begin{cases} 0 & t < 0 \\ 0.15625t^2 - (2.5 \times 10^{-3})t + 10^{-6} & 0 \leq t \leq 6 \text{ ms} \\ 0.625 \times 10^{-6} & t \geq 6 \text{ ms} \end{cases}$$

Answer:

## 4.2 TIME-DEPENDENT SOURCES

Time-dependent periodic waveforms appear frequently in practical applications and are a useful approximation of many physical phenomena. For example, electric power worldwide is generated and delivered to industrial and household users in the form of



**Figure 4.15** Periodic waveforms

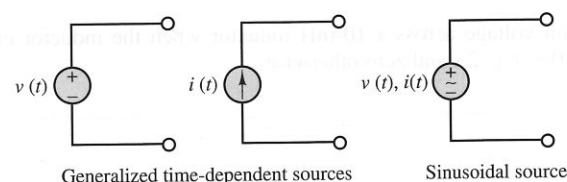
periodic (i.e., 50- or 60-Hz sinusoidal) voltages and currents. The methods developed in this chapter apply to many engineering systems, not just to electric circuits, and will be encountered again in the study of dynamic systems and control systems.

In general, a periodic waveform  $x(t)$  satisfies the equation

$$x(t) = x(t + nT) \quad n = 1, 2, 3, \dots \quad (4.17)$$

where  $T$  is the period of  $x(t)$ . Figure 4.15 illustrates a number of periodic waveforms that are typically encountered in the study of electric circuits. Waveforms such as the sine, triangle, square, pulse, and sawtooth waves are provided in the form of voltages (or, less frequently, currents) by commercially available signal generators.

In this chapter, time-varying voltages and currents and, in particular, sinusoidal (AC) sources are introduced. Figure 4.16 illustrates the convention employed to denote time-dependent sources.



**Figure 4.16** Time-dependent sources

Sinusoids constitute the most important class of time-dependent waveforms. A generalized sinusoid is defined as

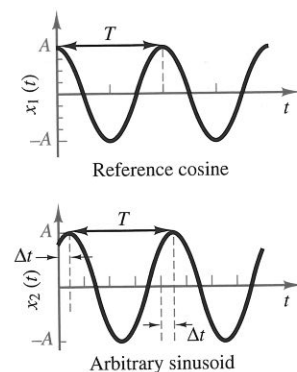
$$x(t) = A \cos(\omega t + \phi) \quad (4.18)$$

where  $A$  is the peak amplitude,  $\omega$  the angular frequency, and  $\phi$  the phase angle. Figure 4.17 summarizes the definitions of  $A$ ,  $\omega$ , and  $\phi$  for the waveforms

$$x_1(t) = A \cos(\omega t) \quad \text{and} \quad x_2(t) = A \cos(\omega t + \phi)$$

where

$$\begin{aligned} f &= \text{cyclical frequency} = \frac{1}{T} \quad \text{cycles/s, or Hz} \\ \omega &= \text{angular frequency} = 2\pi f \quad \text{rad/s} \\ \phi &= 2\pi \frac{\Delta t}{T} \quad \text{rad} \\ &= 360 \frac{\Delta t}{T} \quad \text{deg} \end{aligned} \quad (4.19)$$



**Figure 4.17** Sinusoidal waveforms

The value of the phase shift  $\phi$  is a measure of the time shift of a sinusoid relative to a reference sinusoid, typically a cosine waveform. For example, a sine wave can be represented in terms of a cosine wave by introducing a phase shift of  $\pi/2$  radians:

$$A \sin(\omega t) = A \cos\left(\omega t - \frac{\pi}{2}\right) \quad (4.20)$$

Notice that a negative phase angle represents a time shift to the right.

Although angular frequency  $\omega$ , in units of radians per second, is commonly used to denote sinusoidal frequency, it is also common to employ the cyclical frequency  $f$  in units of cycles per second, or hertz (Hz). In music theory, a sinusoid is a pure tone;

an A-440, for example, is a tone at a frequency of 440 Hz. The cyclical frequency is related to the angular frequency by the factor  $2\pi$ .

$$\omega = 2\pi f \quad \text{Angular frequency} \quad (4.21)$$

### Average (Mean) Value

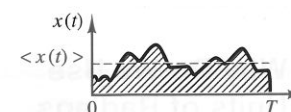
Various measures exist for quantifying the amplitude of a time-varying electric signal. One of these measures is the average or mean value (also called the DC value). The average value of a waveform is computed by integrating it over a suitably chosen period, as shown below.

$$\langle x(t) \rangle = \frac{1}{T} \int_0^T x(\tau) d\tau \quad \text{Average or mean value} \quad (4.22)$$

where  $T$  is the period of integration. Figure 4.18 illustrates the average amplitude of  $x(t)$  over a period of  $T$  seconds. It can be shown that the average or mean value of a sinusoid is zero.

$$\langle A \cos(\omega t + \phi) \rangle = 0 \quad (4.23)$$

This result might be perplexing at first: If any sinusoidal voltage or current has zero average value, is its average power equal to zero? Clearly, the answer must be no. Otherwise, it would be impossible to illuminate households and streets and power industrial machinery with 60-Hz sinusoidal current!



**Figure 4.18** Average of a waveform

### Effective or RMS Value

A more useful measure of the amplitude of an AC waveform  $x(t)$  is the root-mean-square (or rms) value, which takes into account fluctuations of a waveform about its mean, and which is defined as:

$$x_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T x^2(t') dt'} \quad \text{Root-mean-square value} \quad (4.24)$$

Notice that the argument of the square root is the mean value of  $x^2(t)$ . Thus, the RMS value is literally the square root of the mean of the square. Also note that the unit of the “mean of the square” is the unit of  $x^2(t)$ . Thus, the unit of the “root of the mean of the square”  $x_{\text{rms}}$  is the unit of  $x(t)$ .

Why are rms values useful? Consider two similar circuits, each with a resistor  $R$  connected to a source: one with a DC source, and one with an AC source, as shown in Figure 4.19. The effective value of the AC source is the DC source value such that the average power dissipated by the resistor  $R$  is the same in both circuits. Thus, the

effective value of an AC source provides a clear measure of the power associated with the source, which can now be simply expressed as:

$$P_{\text{avg}} = I_{\text{eff}}^2 R = \frac{V_{\text{eff}}^2}{R} \quad (4.25)$$

But how is this discussion of effective values related to rms values? It can be shown that the effective value of an AC source is equivalent to its rms value! That is,

$$I_{\text{eff}} = I_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T i_{\text{ac}}^2(\tau) d\tau} \quad \text{and} \quad V_{\text{eff}} = V_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T v_{\text{ac}}^2(\tau) d\tau} \quad (4.26)$$



The rms, or effective, value of an AC source is the DC value that produces the same average power to be dissipated by a common resistor.

The effective (or rms) value of a voltage or current is indicated by the notation  $V_{\text{rms}}$ , or  $\tilde{V}$ , and  $I_{\text{rms}}$ , or  $\tilde{I}$ . Example 4.9 shows that the ratio of the rms value of a sinusoid to its peak value is  $1/\sqrt{2} \approx 0.707$ . In general, the same ratio for a different type of waveform, such as a square wave, triangle wave, or sawtooth wave, will yield a different value. Table 4.4 lists the value of this ratio for these waveforms. The table also lists a Fourier sine series for each waveform to demonstrate that each is a summation of sine waves.

Table 4.4 Ratio of RMS value to Peak value

| Waveform | $x(t)$                                                                              | $x_{\text{rms}}/x_{\text{pk}}$     |
|----------|-------------------------------------------------------------------------------------|------------------------------------|
| Sinusoid | $A \sin(\omega t)$                                                                  | $\frac{\sqrt{2}}{2} \approx 0.707$ |
| Square   | $\frac{8A}{\pi} \sum_{k=1}^{\infty} \frac{\sin[(2k-1)\omega t]}{2k-1}$              | 1                                  |
| Triangle | $\frac{8A}{\pi^2} \sum_{k=1}^{\infty} (-1)^k \frac{\sin[(2k-1)\omega t]}{(2k-1)^2}$ | $\frac{\sqrt{3}}{3} \approx 0.577$ |
| Sawtooth | $\frac{2A}{\pi} \sum_{k=1}^{\infty} \frac{\sin(k\omega t)}{k}$                      | $\frac{\sqrt{3}}{3} \approx 0.577$ |

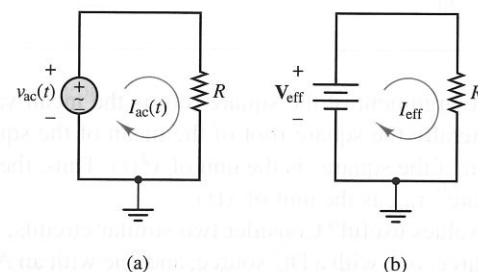


Figure 4.19 AC and DC circuits used to illustrate the concept of effective and rms values

### EXAMPLE 4.8 Average Value of Sinusoidal Waveform



#### Problem

Compute the average value of the signal  $x(t) = 10 \cos(100t)$ .

#### Solution

**Known Quantities:** Functional form of the periodic signal  $x(t)$ .

**Find:** Average value of  $x(t)$ .

**Analysis:** The signal is periodic with period  $T = 2\pi/\omega = 2\pi/100$ ; thus we need to integrate over only one period to compute the average value:

$$\begin{aligned} \langle x(t) \rangle &= \frac{1}{T} \int_0^T x(t') dt' = \frac{100}{2\pi} \int_0^{2\pi/100} 10 \cos(100t) dt \\ &= \frac{10}{2\pi} \langle \sin(2\pi) - \sin(0) \rangle = 0 \end{aligned}$$

**Comments:** The mean value of a sinusoidal is zero, independent of its amplitude and frequency.

### EXAMPLE 4.9 RMS Value of Sinusoidal Waveform



#### Problem

Compute the rms value of the sinusoidal current  $i(t) = I \cos(\omega t)$ .

#### Solution

**Known Quantities:** Functional form of the periodic signal  $i(t)$ .

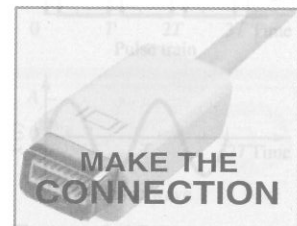
**Find:** RMS value of  $i(t)$ .

**Analysis:** Applying the definition of rms value in equation 4.26, we compute

$$\begin{aligned} i_{\text{rms}} &= \sqrt{\frac{1}{T} \int_0^T i^2(t') dt'} = \sqrt{\frac{\omega}{2\pi} \int_0^{2\pi/\omega} I^2 \cos^2(\omega t') dt'} \\ &= \sqrt{\frac{\omega}{2\pi} \int_0^{2\pi/\omega} I^2 \left[ \frac{1}{2} + \frac{1}{2} \cos(2\omega t') \right] dt'} \\ &= \sqrt{\frac{1}{2} I^2 + \frac{\omega}{2\pi} \int_0^{2\pi/\omega} \frac{I^2}{2} \cos(2\omega t') dt'} \end{aligned}$$

At this point, we recognize that the integral under the square root sign is equal to zero (see Example 4.8), because we are integrating a sinusoidal waveform over two periods. Hence,

$$i_{\text{rms}} = \frac{I}{\sqrt{2}} = 0.707I$$



### Why Do We Use Units of Radians for the Phase Angle $\phi$ ?

The engineer finds it frequently more intuitive to refer to the phase angle in units of degrees; however, to use consistent units in the argument (the quantity in the parentheses) of the expression  $x(t) = A \sin(\omega t + \phi)$ , we must express  $\phi$  in units of radians, since the units of  $\omega t$  are  $[\omega] \cdot [t] = (\text{rad/s}) \cdot \text{s} = \text{rad}$ . Thus, we will consistently use units of radians for the phase angle  $\phi$  in all expressions of the form  $x(t) = A \sin(\omega t + \phi)$ . To be consistent is especially important when one is performing numerical calculations; if one used units of degrees for  $\phi$  in calculating the value of  $x(t) = A \sin(\omega t + \phi)$  at a given  $t$ , the answer would be incorrect.

where  $I$  is the peak value of the waveform  $i(t)$ .

**Comments:** The rms value of a sinusoidal signal is independent of its amplitude and frequency.

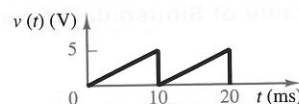
### CHECK YOUR UNDERSTANDING

Express the voltage  $v(t) = 155.6 \sin(377t + \pi/6)$  in cosine form. Note that the angular frequency  $\omega = 377$  rad/s is equivalent to the cyclical frequency 60 Hz, which is the frequency of the electric power generated in North America.

$$\text{Answer: } v(t) = 155.6 \cos(377t - \pi/3)$$

### CHECK YOUR UNDERSTANDING

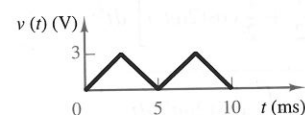
Compute the mean (average) and rms values of the sawtooth waveform shown below.



$$\text{Answers: } v_{\text{avg}} = 2.5 \text{ V}; v_{\text{rms}} = 2.89 \text{ V}$$

### CHECK YOUR UNDERSTANDING

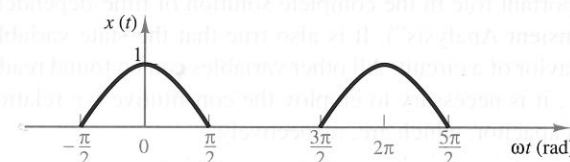
Compute the mean (average) and rms values of the triangle waveform shown below.



$$\text{Answers: } v_{\text{avg}} = 1.5 \text{ V}; v_{\text{rms}} = \sqrt{3} \text{ V}$$

### CHECK YOUR UNDERSTANDING

Compute the mean (average) and rms values of the clipped cosine waveform shown below.



$$x(t) = \cos t \quad \text{for } -\frac{\pi}{2} \leq \omega t < \frac{\pi}{2}$$

$$= 0 \quad \text{for } \frac{\pi}{2} \leq \omega t < \frac{3\pi}{2} \quad \omega = 1$$

$$\text{Answers: } x_{\text{avg}} = 1/\pi; x_{\text{rms}} = 0.5$$

## 4.3 CIRCUITS CONTAINING ENERGY STORAGE ELEMENTS

The resistive circuits studied in Chapters 2 and 3 had no dependence on time. The sources had constant (DC) values and the  $i$ - $v$  relationship for resistors (Ohm's law) had no time dependence. As a result, all the equations obtained in those chapters were algebraic and the voltages and currents were all constants. If a sinusoidal source is present in a resistive circuit, the voltages and currents in the circuit will no longer be constant but instead will vary sinusoidally in time with the same frequency and phase angle as the source. Also, just as with DC circuits, the amplitudes of the voltages and currents will depend upon the resistive network. A circuit with a sinusoidal source is known as an AC circuit.

Purely resistive AC circuits offer no new challenges compared to DC circuits. However, when capacitors and/or inductors are introduced into an AC circuit, the resulting behavior is significantly more interesting and challenging. The reason is that the  $i$ - $v$  relationships for capacitors and inductors are time dependent. The result, in general, is that the amplitudes and phase angles of voltages and currents in the circuit can be different from those of the source. Consequently, in the solution of AC circuits it is necessary to keep track of two variables (amplitude and phase) for each voltage and current. By contrast, when solving DC circuits, it was necessary to keep track of only one variable (amplitude). Note that the frequency of all voltages and currents in an AC circuit equals the source frequency.

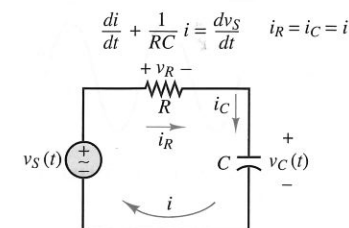
To clarify this discussion, consider the simple series loop shown in Figure 4.20, which consists of a known sinusoidal voltage source, a resistor, and a capacitor. Apply KVL around the loop to obtain the governing equation:

$$v_S - v_R - v_C = 0 \quad \text{or} \quad v_R + v_C = v_S \quad (4.27)$$

The so-called *state variable* for this circuit is the voltage  $v_C$  across the capacitor. The state variables in a circuit are the voltages across capacitors and the currents through inductors. Note that capacitors in parallel share the same state variable. The same is true for inductors in series. Capacitors in series and inductors in parallel can also be



A circuit containing energy storage elements is described by a differential equation.



**Figure 4.20** Circuit containing energy storage element

reduced to a single equivalent capacitance and inductance, respectively, with a single state variable for each equivalent capacitance and inductance. As a result, the number of state variables in a circuit is equal to the number of irreducible capacitors and inductors. In general, it is a good idea to learn to solve for state variables first because they play an important role in the complete solution of time-dependent circuits (see Chapter 5, “Transient Analysis”). It is also true that the state variables completely describe the behavior of a circuit. All other variables can be found readily from them.

To find  $v_C$ , it is necessary to employ the constitutive  $i$ - $v$  relationships for the resistor and the capacitor, which are, respectively:

$$v_R = i_R R \quad \text{and} \quad i_C = C \frac{dv_C}{dt} \quad (4.28)$$

Notice that the resistor current and the capacitor current are the same for this simple loop. Thus:

$$v_R = i_R R = i_C R = RC \frac{dv_C}{dt} \quad (4.29)$$

Plug this result into Equation 4.27 to obtain:

$$RC \frac{dv_C}{dt} + v_C = v_S \quad (4.30)$$

Divide both sides of Equation 4.30 to find the standard form:

$$\frac{dv_C}{dt} + \frac{1}{RC} v_C = \frac{1}{RC} v_S \quad (4.31)$$

The result is a first-order, linear, ordinary differential equation. The solution for  $v_C$  has two parts, as usual: (1) a transient solution, and (2) a steady-state solution. The complete solution of the differential equation is the sum of these two parts. It is important to note that once the complete solution for  $v_C$  is found, it is a simple matter to find  $i(t)$  and  $v_R(t)$  from equations 4.28 and 4.29, respectively.

It is also possible to find similar differential equations in the variables  $i(t)$  and  $v_R(t)$ . For  $i(t)$ , the result is:

$$\frac{di}{dt} + \frac{1}{RC} i = \frac{1}{R} \frac{dv_S}{dt} \quad (4.32)$$

Notice that the left-hand side of this equation is identical to that found in equation 4.31. Only the right-hand side is different. The differential equation for  $v_R(t)$  follows the same rule. The constant  $RC$  has units of time and is a common example of an important class of parameters known as time constants.

For more complicated circuits, the process is largely the same except that KVL and KCL may have to be applied multiple times and the circuit may contain multiple resistors, capacitors, and inductors. The result will be multiple first- and perhaps second-order linear, ordinary differential equations. It is not difficult to imagine that for even modest circuits the procedure and results may become quite complicated and cumbersome. In fact, in some cases the above procedure would effectively require that all state variables in a circuit be solved simultaneously, when only one particular voltage or current may be sought.

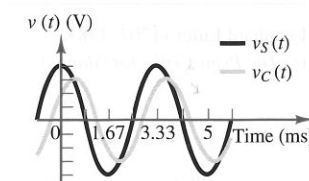
To avoid these complications, an alternative approach is to dispense with time derivatives as often as possible and solve for the steady-state (particular) and transient (homogeneous) solutions separately using the following two methods:

- **Steady-state solution.** To solve for the steady-state solution, Euler's formula is employed to represent sinusoids as complex exponentials and to eliminate the time derivatives in the constitutive  $i$ - $v$  relations for capacitors and inductors. The result is algebraic equations with complex constants and variables. These equations can be solved using standard algebra techniques. The only complication is that the arithmetic involves complex numbers rather than real numbers. As an added bonus, many of the intermediate details of this approach need only be understood but not actually executed during the solution of a circuit problem.

- **Transient solution.** Whenever possible, Thévenin's and Norton's theorems are used to simplify complicated circuits and to focus on solving for the state variables. The result will often be simple first- or second-order circuit archetypes with previously established solutions, such that no formal solving of differential equations is necessary. This approach is beneficial whenever all capacitors and inductors can be isolated within a relatively simple load. The remainder of the circuit will then be comprised of resistors and independent sources and can be simplified using techniques explored in Chapter 3. The transient solution is explored in greater detail in Chapter 5.

The remainder of this chapter is focused on steady-state solutions of circuits like the one in Figure 4.20 and others more complicated. Although these steady-state solutions do not contain sinusoidal functions explicitly, it is important to keep in mind that they nonetheless represent sinusoids and can be converted to explicit sinusoids using Euler's formula in the end. A typical depiction of these sinusoidal solutions is shown in Figure 4.21. The dark black curve represents the sinusoidal voltage source that is driving the circuit in Figure 4.20. The gray curve represents the sinusoidal voltage across the capacitor responding to the driving source. Notice that the capacitor voltage is a scaled version of the source and is time shifted (i.e., phase shifted) with respect to it. The driving source acts as the reference against which all other voltages and currents are compared. The results shown in Figure 4.21 are typical of a steady-state solution and can be summarized as follows:

In a linear circuit with a sinusoidal source, all voltages and currents are sinusoids at the same frequency as the source. These voltages and currents are *scaled* versions of the source and may be shifted in time (i.e., phase shifted) with respect to it.



**Figure 4.21** Waveforms for the AC circuit of Figure 4.20

## CHECK YOUR UNDERSTANDING

Find the differential equation for  $v_R(t)$  for the circuit in Figure 4.20.

Answer:  $\frac{dv_R}{dt} = \frac{RC}{1} v_R + \frac{dv_S}{dt}$

#### 4.4 PHASOR SOLUTION OF CIRCUITS WITH SINUSOIDAL SOURCES

In this section, an efficient notation is introduced to represent sinusoidal signals as complex numbers and to eliminate the need for solving differential equations. (A reasonably complete treatment of complex algebra is presented in Appendix A, including examples and exercises.) The remainder of the chapter assumes that the reader is familiar with both rectangular and polar forms of complex numbers; with the conversion between these two forms; and with the basic operations of addition, subtraction, multiplication, and division of complex numbers.

Any sinusoidal signal may be represented either in the time domain:

$$v(t) = A \cos(\omega t + \theta)$$

or in the frequency domain (also known as the phasor) form:

$$V(j\omega) = Ae^{j\theta} = A\angle\theta = A(\cos\theta + j\sin\theta)$$

The argument  $j\omega$  indicates the  $e^{j\omega t}$  time-dependence of the phasor.

A phasor is a complex number consisting of a magnitude equal to the peak amplitude of a sinusoid and a phase angle equal to the phase shift of the same sinusoid with respect to a reference sinusoid, usually a source. A phase shift in the frequency domain is equivalent to a time delay in the time domain.

Since the sinusoidal source frequency  $\omega$  is common to all phasors in an AC circuit, the complex exponential  $e^{j\omega t}$  is not expressed explicitly. Thus, it is important to note the specific frequency  $\omega$  of each sinusoidal source in a circuit.



Leonhard Euler (1707–1783)  
(© The Print Collector/Alamy.)

##### Euler's Formula

Named after the famous Swiss mathematician Leonhard Euler, this formula is the basis of phasor notation. A phasor is similar to a vector in that it has an amplitude and direction in the complex plane. Also, just as a vector can be decomposed into  $x$  and  $y$  components, a phasor can be decomposed into real and imaginary components. Euler's formula defines a complex exponential  $e^{j\theta}$  as a unit phasor in the complex plane, with real and imaginary components given by:

$$e^{j\theta} = \cos\theta + j\sin\theta \quad (4.33)$$

where  $j \equiv \sqrt{-1}$  is the imaginary unit. The symbol  $\theta$  is simply a place holder in Euler's formula. Any quantity or expression can be substituted for  $\theta$  in the formula. In the next section on phasors,  $\theta$  takes on the physical meaning of the phase shift of a sinusoid.

The **dark** black arrow in Figure 4.22 represents the complex exponential in the complex plane. The real and imaginary components of the complex exponential are shown as  $\cos\theta$  and  $\sin\theta$ , respectively. These two components and the complex exponential itself form the three legs of a right triangle. The Pythagorean theorem requires:

$$|e^{j\theta}|^2 = \cos^2\theta + \sin^2\theta = 1 \quad (4.34)$$

Thus, the magnitude of  $e^{j\theta}$  is unity, which is why it is also known as a unit phasor. The angle of inclination of the unit phasor is  $\theta$ . As  $\theta$  increases or decreases the unit phasor rotates counterclockwise or clockwise, respectively, about the origin of the complex plane.

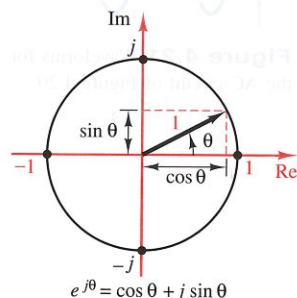


Figure 4.22 Euler's formula

It is difficult to overstate the power of the visualization presented in Figure 4.22. For example, when  $\theta = \pi/2$ , the unit phasor points straight up along the imaginary axis. Thus:

$$e^{j\pi/2} = 1\angle\frac{\pi}{2} = j \quad (4.35)$$

where the notation  $1\angle\frac{\pi}{2}$  indicates a magnitude of 1 and a phase angle  $\theta = \pi/2$ . When  $\theta = \pi$ , the unit phasor points to the left along the negative real axis. Thus:

$$e^{j\pi} = 1\angle\pi = -1 \quad (4.36)$$

Likewise:

$$e^{j3\pi/2} = 1\angle\frac{3\pi}{2} = -j \quad \text{and} \quad e^{j2\pi} = 1\angle2\pi = 1 \quad (4.37)$$

Each of these expressions equates the two polar forms on the left to the rectangular form on the far right side. In polar form, a phasor is represented by a magnitude (or amplitude) and a phase angle, whether as  $Ae^{j\theta}$  or  $A\angle\theta$ . In rectangular form, a phasor is represented by real and imaginary components. Table 4.5 lists a few other commonly encountered phasors in polar and rectangular forms.

Table 4.5 Polar and rectangular forms of common phasors

| Complex exponential      | Polar                    | Rectangular                     |
|--------------------------|--------------------------|---------------------------------|
| $Ae^{\pm j(\pi/6)}$      | $A\angle\pm\pi/6$        | $A(\sqrt{3}/2 \pm j/2)$         |
| $Ae^{\pm j(\pi/4)}$      | $A\angle\pm\pi/4$        | $A(\sqrt{2}/2 \pm j\sqrt{2}/2)$ |
| $Ae^{\pm j(\pi/3)}$      | $A\angle\pm\pi/3$        | $A(1/2 \pm j\sqrt{3}/2)$        |
| $Ae^{\pm j\arctan(3/4)}$ | $A\angle\pm\arctan(3/4)$ | $A(0.8 \pm j0.6)$               |
| $Ae^{\pm j\arctan(4/3)}$ | $A\angle\pm\arctan(4/3)$ | $A(0.6 \pm j0.8)$               |

In general, the polar and rectangular forms are related by:

$$Ae^{j\theta} = A\angle\theta = A\cos\theta + jA\sin\theta \quad (4.38)$$

In effect, Euler's identity is simply a trigonometric relationship in the complex plane.

##### Phasors

To see how complex numbers can be used to represent sinusoidal signals, rewrite the expression for a generalized sinusoid in light of Euler's equation:

$$A\cos(\omega t + \theta) = \text{Re}\{Ae^{j(\omega t + \theta)}\} \quad (4.39)$$

Notice that it is possible to express any sinusoid as the real part of a complex exponential with an argument of  $\omega t + \theta$  and a magnitude or amplitude of  $A$ . The expression can be further simplified by remembering that the angular frequency  $\omega$  is common to all voltages and currents. Thus, the  $e^{j\omega t}$  portion of the complex exponential is understood to be present in every phasor, but not written explicitly. The same perspective is taken with regard to the real part operator  $\text{Re}$  so that the complex exponential is simplified as shown below.

$$\text{Re}\{Ae^{j(\omega t + \theta)}\} = \text{Re}\{Ae^{j\omega t}e^{j\theta}\} \Rightarrow Ae^{j\theta} \quad (4.40)$$

In this expression, the relational operator  $\Rightarrow$  indicates equality but with the real part operator  $\text{Re}$  and the sinusoidal portion  $e^{j\omega t}$  of the complex exponential hidden but

understood implicitly. In general, this simplification will be used to express a phasor in polar and rectangular form as:

$$Ae^{j\theta} = A\angle\theta = A(\cos\theta + j\sin\theta) \quad \text{Phasor notation} \quad (4.41)$$

The reason for this simplification is convenience, as will become apparent in the examples. It is imperative to remember that the  $e^{j\omega t}$  term is still present implicitly.

In section 4.5, a new quantity known as *impedance* is introduced. Impedance is defined as the ratio of a voltage phasor to a current phasor. At this point, it is worth mentioning five key rules of complex arithmetic that will be used to resolve complex multiplication and division:

1. The magnitude of the ratio of two phasors is the ratio of the individual magnitudes. For example,  $|\mathbf{V}/\mathbf{I}| = |\mathbf{V}|/|\mathbf{I}|$ .
2. The phase angle of the ratio of two phasors is the difference of the individual phase angles. For example,  $\angle(\mathbf{V}/\mathbf{I}) = \angle(\mathbf{V}) - \angle(\mathbf{I})$ .
3. The complex conjugate  $\bar{\mathbf{A}}$  of a phasor  $\mathbf{A}$  is found by changing the sign of the imaginary unit,  $j$ , everywhere in the phasor. The magnitude of the complex conjugate of a phasor equals the magnitude of the phasor itself. The angle of the complex conjugate of a phasor equals the negative of the angle of the phasor itself.
4. The product of a phasor and its complex conjugate is a real number equal to the square of the magnitude of the phasor, which is equal to the sum of the square of the real part of the phasor and the square of the imaginary part of the phasor.
5. The angle of a phasor is the inverse tangent of the ratio of the imaginary part to the real part. That is,  $\angle\mathbf{A} = \arctan([\text{Im}(\mathbf{A})/\text{Re}(\mathbf{A})])$ .

A bold uppercase font indicates a phasor quantity.

### Superposition of AC Signals

Example 4.10 explores the effect of having two sinusoidal sources of different phase and amplitude, but of the same frequency, in a circuit. It is important to realize that the approach used there *does not apply* to the superposition of two (or more) sinusoidal sources that *are not at the same frequency*. The more general case of two sinusoidal sources of two different frequencies is explored here.

Consider the circuit depicted in Figure 4.23 with a load excited by two current sources in parallel.

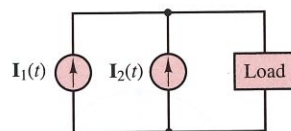


Figure 4.23 Superposition of AC

$$i_1(t) = A_1 \cos(\omega_1 t + \theta_1) \quad (4.42)$$

$$i_2(t) = A_2 \cos(\omega_2 t + \theta_2)$$

By KCL, the load current is equal to the sum of the two source currents; that is,

$$i_{\text{load}}(t) = i_1(t) + i_2(t) \quad (4.43)$$

So far, so good. However, the expression in Equation 4.43 *cannot* be expressed in phasor form without masking the fact that  $i_1$  has a different frequency than that of  $i_2$ . For example, it may be tempting to write:

$$\begin{aligned} \mathbf{I}_{\text{load}} &= \mathbf{I}_1 + \mathbf{I}_2 \\ &= A_1 e^{j\theta_1} + A_2 e^{j\theta_2} \end{aligned} \quad (4.44)$$

However, it is imperative to remember that the  $e^{j\omega_1 t}$  and  $e^{j\omega_2 t}$  terms are present implicitly in  $\mathbf{I}_1$  and  $\mathbf{I}_2$ , respectively, as shown below.

$$\begin{aligned} i_1(t) &= \text{Re}(\mathbf{I}_1 e^{j\omega_1 t}) \\ i_2(t) &= \text{Re}(\mathbf{I}_2 e^{j\omega_2 t}) \end{aligned} \quad (4.45)$$

The two phasors of equation 4.44 cannot be added, but must be kept separate; the only unambiguous expression for the load current is equation 4.43. To analyze a circuit with multiple sinusoidal sources at different frequencies, it is necessary to solve the circuit separately for each source and add the individual answers. Example 4.11 illustrates the use of AC superposition to determine the response of a circuit with two sources of different frequencies.

### 4.5 IMPEDANCE

The  $i$ - $v$  relationships of resistors, capacitors, and inductors can be expressed in phasor notation. As phasors, each  $i$ - $v$  relationship takes the form of a generalized Ohm's law:

$$\mathbf{V} = \mathbf{I}\mathbf{Z}$$

where the phasor quantity  $\mathbf{Z}$  is known as impedance. For a resistor, inductor, and capacitor, the impedances are, respectively:

$$\mathbf{Z}_R = R \quad \mathbf{Z}_L = j\omega L \quad \mathbf{Z}_C = \frac{1}{j\omega C} = \frac{-j}{\omega C}$$

Combinations of resistors, inductors, and capacitance can be represented by a single equivalent impedance of the form:

$$\mathbf{Z}(j\omega) = R(j\omega) + jX(j\omega) \quad \text{units of } \Omega \text{ (ohms)}$$

where  $R(j\omega)$  and  $X(j\omega)$  are known as the “resistance” and “reactance” portions, respectively, of the equivalent impedance  $\mathbf{Z}$ . Both terms are, in general, functions of frequency  $\omega$ .

The admittance is defined as the inverse of impedance.

$$\mathbf{Y} \equiv \frac{1}{\mathbf{Z}} \quad \text{units of S (siemens)}$$

Consequently, all the DC circuit relations and techniques introduced in Chapter 3 can be extended to AC circuits. Thus, it is not necessary to learn new techniques and formulas to solve AC circuits; it is only necessary to learn to use the same techniques and formulas with phasors.

### Generalized Ohm's Law

The impedance concept reflects the fact that capacitors and inductors act as frequency-dependent resistors. Figure 4.24 depicts a generic AC circuit with a sinusoidal voltage source  $\mathbf{V}_S$  phasor and an impedance load  $\mathbf{Z}$ , which is also a phasor and represents the effect of a generic network of resistors, capacitors, and inductors. The resulting current  $\mathbf{I}$  is a phasor determined by:

$$\mathbf{V} = \mathbf{I}\mathbf{Z} \quad \text{Generalized Ohm's law} \quad (4.46)$$

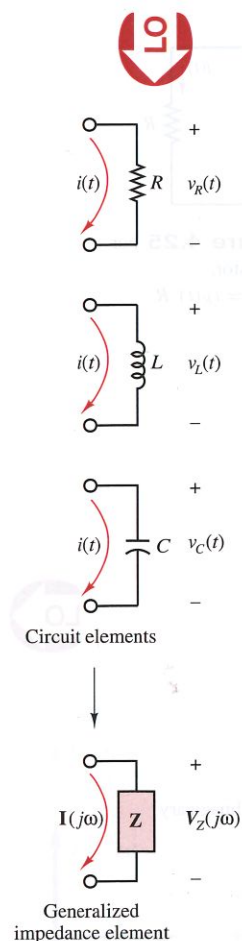


Figure 4.24 The impedance concept

A specific expression for the impedance  $\mathbf{Z}$  is found for each specific network of resistors, capacitors, and inductors attached to the source. To determine  $\mathbf{Z}$  it is first necessary to determine the impedance of resistors, capacitors, and inductors using the definition of impedance:



$$\mathbf{Z} \equiv \frac{\mathbf{V}}{\mathbf{I}} \quad \text{Definition of impedance} \quad (4.47)$$

Once the impedance of each resistor, capacitor, and inductor in a network is known, they can be combined in series and parallel (using the usual rules for resistors) to form an equivalent impedance “seen” by the source.

### Impedance of a Resistor

The  $i$ - $v$  relationship for a resistor is, of course, Ohm’s law, which in the case of sinusoidal sources is written as (see Figure 4.25):

$$v_R(t) = i_R(t)R \quad \text{or, in phasor form,} \quad (4.48)$$

$$\mathbf{V}_R e^{j\omega t} = \mathbf{I}_R e^{j\omega t} R$$

where  $\mathbf{V}_R = V_R e^{j\theta_v}$  and  $\mathbf{I}_R = I_R e^{j\theta_i}$  are phasors.

Both sides of equation 4.47 can be divided by  $e^{j\omega t}$  to yield:

$$\mathbf{V}_R = \mathbf{I}_R R \quad (4.49)$$

The impedance of a resistor is then determined from the definition of impedance:

$$\mathbf{Z}_R \equiv \frac{\mathbf{V}_R}{\mathbf{I}_R} = R \quad (4.50)$$

Thus:



$$\mathbf{Z}_R = R \quad \text{Impedance of a resistor} \quad (4.51)$$

The impedance of a resistor is a real number; that is, it has a magnitude  $R$  and a zero phase, as shown in Figure 4.26. The phase of the impedance is equal to the phase difference between the voltage across an element and the current through the same element. In the case of a resistor, the voltage is completely in phase with the current, which means that there is no time delay or time shift between the voltage waveform and the current waveform in the time domain.

It is important to keep in mind that the phasor voltages and currents in AC circuits are functions of frequency,  $\mathbf{V} = \mathbf{V}(j\omega)$  and  $\mathbf{I} = \mathbf{I}(j\omega)$ . This fact is crucial for determining the impedance of capacitors and inductors, as shown below.

### Impedance of an Inductor

The  $i$ - $v$  relationship for an inductor is (see Figure 4.27):

$$v_L(t) = L \frac{di_L(t)}{dt} \quad (4.52)$$

At this point, it is important to proceed carefully. The time-domain expression for the current through the inductor is:

$$i_L(t) = I_L \cos(\omega t + \theta)$$

such that

$$\begin{aligned} \frac{d}{dt} i_L(t) &= -I_L \omega \sin(\omega t + \theta) \\ &= I_L \omega \cos(\omega t + \theta + \pi/2) \\ &= \text{Re} (I_L \omega e^{j\pi/2} e^{j\omega t + \theta}) \\ &= \text{Re} [I_L(j\omega) e^{j\omega t + \theta}] \end{aligned} \quad (4.53)$$

Notice that the net effect of the time derivative is to produce an extra  $(j\omega)$  term along with the complex exponential expression of  $i_L(t)$ . That is:

| Time domain    | Frequency domain |
|----------------|------------------|
| $\frac{d}{dt}$ | $j\omega$        |

Therefore, the phasor equivalent of the  $i$ - $v$  relationship for an inductor is:

$$\mathbf{V}_L = L(j\omega) \mathbf{I}_L \quad (4.54)$$

The impedance of an inductor is then determined from the definition of impedance:

$$\mathbf{Z}_L \equiv \frac{\mathbf{V}_L}{\mathbf{I}_L} = j\omega L \quad (4.55)$$

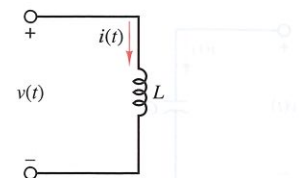
Thus:



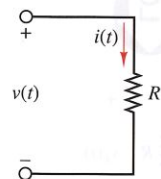
$$\mathbf{Z}_L = j\omega L = \omega L \angle \frac{\pi}{2} \quad \text{Impedance of an inductor} \quad (4.56)$$

The impedance of an inductor is a positive, purely imaginary number; that is, it has a magnitude of  $\omega L$  and a phase of  $\pi/2$  radians or  $90^\circ$ , as shown in Figure 4.28. As before, the phase of the impedance is equal to the phase difference between the voltage across an element and the current through the same element. In the case of an inductor, the voltage leads the current by  $\pi/2$  radians, which means that a feature (e.g., a zero crossing point) of the voltage waveform occurs  $T/4$  seconds earlier than the same feature of the current waveform.  $T$  is the common period.

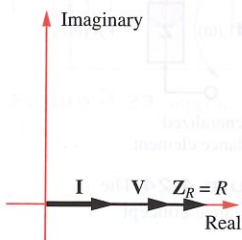
Note that the inductor behaves as a complex frequency-dependent resistor and that its magnitude  $\omega L$  is proportional to the angular frequency  $\omega$ . Thus, an inductor will “impede” current flow in proportion to the frequency of the source signal. At low frequencies, an inductor acts like a short-circuit; at high frequencies, it acts like an open-circuit.



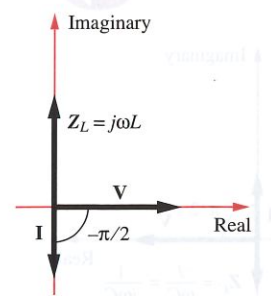
**Figure 4.27** For an inductor,  $v_L(t) = L \frac{di_L(t)}{dt}$



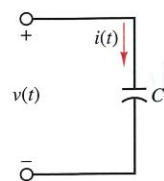
**Figure 4.25** For a resistor,  $v_R(t) = i_R(t)R$



**Figure 4.26** Phasor diagram of the impedance of a resistor. Remember that  $\mathbf{Z} = \mathbf{V}/\mathbf{I}$ .



**Figure 4.28** Phasor diagram of the impedance of an inductor. Remember that  $\mathbf{Z} = \mathbf{V}/\mathbf{I}$ .



**Figure 4.29** For a capacitor,  $i_C(t) = C \frac{d}{dt} v_C(t)$

### Impedance of a Capacitor

The principle of duality suggests that the procedure to derive the impedance of a capacitor should be a mirror image of the procedure shown above for an inductor. The  $i$ - $v$  relationship for a capacitor is (see Figure 4.29):

$$i_C(t) = C \frac{dv_C(t)}{dt} \quad (4.57)$$

The time-domain expression for the voltage across the capacitor is:

$$v_C(t) = V_C \cos(\omega t + \theta)$$

such that

$$\begin{aligned} \frac{d}{dt} v_C(t) &= -V_C \omega \sin(\omega t + \theta) \\ &= V_C \omega \cos(\omega t + \theta + \pi/2) \\ &= \text{Re} (V_C \omega e^{j\pi/2} e^{j\omega t + \theta}) \\ &= \text{Re} [V_C(j\omega) e^{j\omega t + \theta}] \end{aligned} \quad (4.58)$$

Notice that the *net* effect of the time derivative is to produce an extra ( $j\omega$ ) term along with the complex exponential expression of  $v_C(t)$ . Therefore, the phasor equivalent of the  $i$ - $v$  relationship for a capacitor is:

$$\mathbf{I}_C = C(j\omega)\mathbf{V}_C \quad (4.59)$$

The impedance of an inductor is then determined from the definition of impedance:

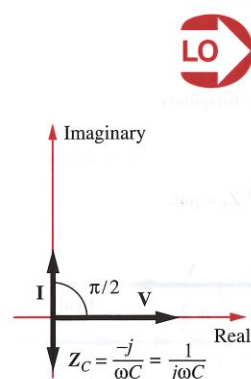
$$\mathbf{Z}_C \equiv \frac{\mathbf{V}_C}{\mathbf{I}_C} = \frac{1}{j\omega C} = \frac{-j}{\omega C} \quad (4.60)$$

Thus:

$$\mathbf{Z}_C = \frac{1}{j\omega C} = \frac{-j}{\omega C} = \frac{1}{\omega C} \angle \frac{-\pi}{2} \quad \text{Impedance of a capacitor} \quad (4.61)$$

The impedance of a capacitor is a negative, purely imaginary number; that is, it has a magnitude of  $1/\omega C$  and a phase of  $-\pi/2$  radians or  $-90^\circ$ , as shown in Figure 4.30. As before, the phase of the impedance is equal to the phase difference between the voltage across an element and the current through the same element. In the case of a capacitor, the voltage *lags* the current by  $\pi/2$  radians, which means that a feature (e.g., a zero crossing point) of the voltage waveform occurs  $T/4$  seconds *later* than the same feature of the current waveform.  $T$  is the common period of each waveform.

Note that the capacitor also behaves as a complex frequency-dependent resistor, except that its magnitude  $1/\omega C$  is inversely proportional to the angular frequency  $\omega$ .



**Figure 4.30** Phasor diagram of the impedance of a capacitor. Remember that  $\mathbf{Z} = \mathbf{V}/\mathbf{I}$ .

Thus, a capacitor will “impede” current flow in inverse proportion to the frequency of the source. At low frequencies, a capacitor acts like an open-circuit; at high frequencies, it acts like a short-circuit.

### Generalized Impedance

The impedance concept is very useful in solving AC circuit analysis problems. It allows network theorems developed for DC circuits to be applied to AC circuits. Examples 4.12 to 4.14 illustrate how circuits with impedance elements in series and parallel are reduced to a single equivalent impedance, in much the same way as was done in resistive circuits. The only difference is that complex arithmetic, rather than scalar arithmetic, must be employed to find the equivalent impedance.

Figure 4.31 depicts  $\mathbf{Z}_R(j\omega)$ ,  $\mathbf{Z}_L(j\omega)$ , and  $\mathbf{Z}_C(j\omega)$  in the complex plane. It is important to emphasize that although the impedance of resistors is purely real and the impedance of capacitors and inductors is purely imaginary, the equivalent impedance seen by a source in an arbitrary circuit can be complex.

$$\mathbf{Z}(j\omega) = R + X(j\omega) \quad (4.62)$$

Here,  $R$  is *resistance* and  $X$  is *reactance*. The unit of  $R$ ,  $X$ , and  $\mathbf{Z}$  is the ohm.

### Admittance

In Chapter 3, it was suggested that the solution of certain circuit analysis problems was handled more easily in terms of conductances than resistances. This is true, for example, when one is using node analysis, or in circuits with many parallel elements, since conductances in parallel add as resistors in series do. In AC circuit analysis, an analogous quantity may be defined—the reciprocal of complex impedance. Just as conductance  $G$  was defined as the inverse of resistance, admittance  $\mathbf{Y}$  is defined as the inverse of impedance.

$$\mathbf{Y} \equiv \frac{1}{\mathbf{Z}} \quad \text{units of S (siemens)} \quad (4.63)$$

Whenever the impedance  $\mathbf{Z}$  is purely real, the admittance  $\mathbf{Y}$  is identical to the conductance  $G$ . In general, however,  $\mathbf{Y}$  is complex.

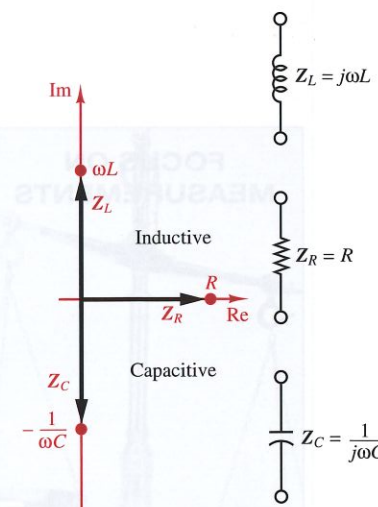
$$\mathbf{Y} = G + jB \quad (4.64)$$

where  $G$  is the AC conductance and  $B$  is the susceptance, which is analogous to reactance. Clearly,  $G$  and  $B$  are related to  $R$  and  $X$ ; however, the relationship is not a simple inverse. If  $\mathbf{Z} = R + jX$ , then the admittance is:

$$\mathbf{Y} = \frac{1}{\mathbf{Z}} = \frac{1}{R + jX} \quad (4.65)$$

Multiply the numerator and denominator by the complex conjugate  $\bar{\mathbf{Z}} = R - jX$ :

$$\mathbf{Y} = \frac{\bar{\mathbf{Z}}}{\mathbf{Z}\bar{\mathbf{Z}}} = \frac{R - jX}{R^2 + X^2} \quad (4.66)$$



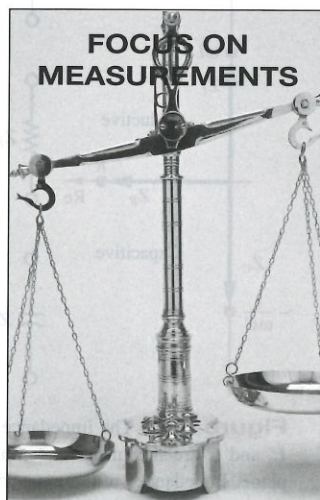
**Figure 4.31** The impedances of  $R$ ,  $L$ , and  $C$  are shown in the complex plane. Impedances in the upper right quadrant are inductive while those in the lower right quadrant are capacitive.

and conclude that

$$\begin{aligned} G &= \frac{R}{R^2 + X^2} \\ B &= \frac{-X}{R^2 + X^2} \end{aligned} \quad (4.67)$$

Notice in particular that  $G$  is not the reciprocal of  $R$  in the general case! Example 4.15 illustrates the determination of  $Y$  for some common circuits.

### FOCUS ON MEASUREMENTS



#### Capacitive Displacement Transducer

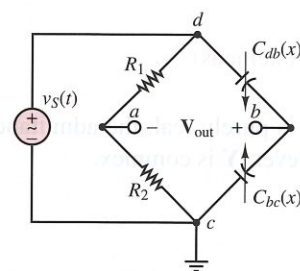
As introduced in the previous Focus on Measurements section, a capacitive displacement transducer consists of a parallel-plate capacitor with a variable separation distance  $x$ . The capacitance was shown to be:

$$C = \frac{8.854 \times 10^{-3} A}{x} \text{ pF}$$

where  $C$  is the capacitance in picofarads,  $A$  is the area of the plates in square millimeters, and  $x$  is the (variable) distance in millimeters. The impedance of the capacitor is:

$$Z_C = \frac{1}{j\omega C} = \frac{x}{j\omega(8.854 \times 10^{-3})A} \text{ T}\Omega$$

Thus, at a given frequency  $\omega$ , the impedance of the capacitor varies linearly with the separation distance. This result can be exploited in a bridge circuit, as shown in Figure 4.32 where half of the bridge is a differential pressure transducer in which a thin diaphragm (plate) is situated between two fixed plates and subject to variations in pressure across the diaphragm. The result is that when the capacitance of one leg of the bridge, shown here again as Figure 4.32, increases, the capacitance of the other leg decreases. Assume the bridge is excited by a sinusoidal source.



**Figure 4.32** Bridge circuit for capacitive displacement transducer

Apply voltage division and KVL to express the output voltage in phasor notation as:

$$V_{out}(j\omega) = V_S(j\omega) \left( \frac{Z_{Cbc}(x)}{Z_{Cdb}(x) + Z_{Cbc}(x)} - \frac{R_2}{R_1 + R_2} \right)$$

(Continued)

(Concluded)

When the diaphragm is not displaced from its center position, the nominal capacitance of each half of the transducer is given by:

$$C = \frac{\epsilon A}{d}$$

where  $d$  is the nominal separation distance between the diaphragm and the fixed surfaces (in millimeters). Thus, when the diaphragm is displaced an effective distance  $x$ , the capacitance of each leg of the bridge is given by:

$$C_{db} = \frac{\epsilon A}{d - x} \quad \text{and} \quad C_{bc} = \frac{\epsilon A}{d + x}$$

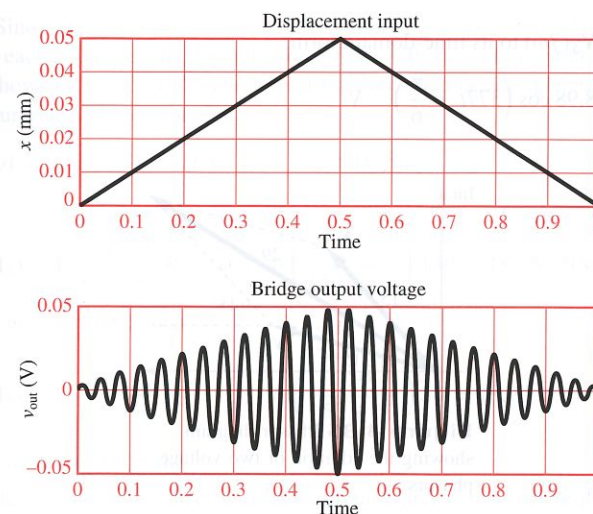
Therefore, the corresponding impedance of each leg is:

$$Z_{Cdb} = \frac{d - x}{j\omega(8.854 \times 10^{-3})A} \quad \text{and} \quad Z_{Cbc} = \frac{d + x}{j\omega(8.854 \times 10^{-3})A}$$

such that the phasor output voltage is:

$$\begin{aligned} V_{out}(j\omega) &= V_S(j\omega) \left( \frac{\frac{d + x}{j\omega(8.854 \times 10^{-3})A}}{\frac{d - x}{j\omega(8.854 \times 10^{-3})A} + \frac{d + x}{j\omega(8.854 \times 10^{-3})A}} - \frac{R_2}{R_1 + R_2} \right) \\ &= V_S(j\omega) \left( \frac{1}{2} + \frac{x}{2d} - \frac{R_2}{R_1 + R_2} \right) \\ &= V_S(j\omega) \frac{x}{2d} \quad (\text{assuming } R_1 = R_2) \end{aligned}$$

Thus, the output voltage will vary as a scaled version of the input voltage in proportion to the displacement. A typical  $v_{out}(t)$  is displayed in Figure 4.33 for a 0.05-mm “triangular” diaphragm displacement, with  $d = 0.5$  mm and  $V_S$  a 25-Hz sinusoid with 1-V amplitude.



**Figure 4.33** Displacement input and bridge output voltage for capacitive displacement transducer



### EXAMPLE 4.10 Addition of Two Sinusoidal Sources Using Phasor Notation

#### Problem

Compute the phasor voltage across a series connection of two sinusoidal voltage sources (Figure 4.34).

#### Solution

##### Known Quantities:

$$v_1(t) = 15 \cos\left(377t + \frac{\pi}{4}\right) \text{ V}$$

$$v_2(t) = 15 \cos\left(377t + \frac{\pi}{12}\right) \text{ V}$$

**Find:** Equivalent phasor voltage  $v_S(t)$ .

**Analysis:** Write the two voltages in phasor form:

$$\mathbf{V}_1(j\omega) = 15\angle\frac{\pi}{4} \text{ V}$$

$$\mathbf{V}_2(j\omega) = 15e^{j\pi/12} = 15\angle\frac{\pi}{12} \text{ V}$$

The phasor diagram of Figure 4.35 shows  $\mathbf{V}_1$  and  $\mathbf{V}_2$  in the complex plane. Convert the phasor voltages from polar to rectangular form:

$$\mathbf{V}_1(j\omega) = 10.61 + j10.61 \text{ V}$$

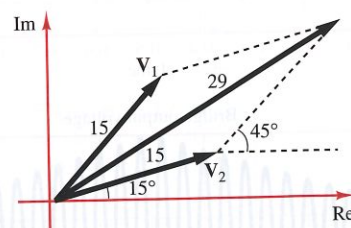
$$\mathbf{V}_2(j\omega) = 14.49 + j3.88 \text{ V}$$

Then, by KVL:

$$\mathbf{V}_S(j\omega) = \mathbf{V}_1(j\omega) + \mathbf{V}_2(j\omega) = 25.10 + j14.49 = 28.98e^{j\pi/6} = 28.98\angle\frac{\pi}{6} \text{ V}$$

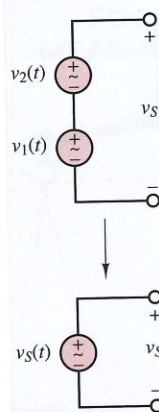
Finally, convert  $\mathbf{V}_S(j\omega)$  to its time-domain form:

$$v_S(t) = 28.98 \cos\left(377t + \frac{\pi}{6}\right) \text{ V}$$



**Figure 4.35** Phasor diagram showing the addition of two voltage phasors.

**Comments:** The same result could have been obtained by adding the two sinusoids in the time domain, using trigonometric identities:



**Figure 4.34**

$$v_1(t) = 15 \cos\left(377t + \frac{\pi}{4}\right) = 15 \cos\frac{\pi}{4} \cos(377t) - 15 \sin\frac{\pi}{4} \sin(377t) \text{ V}$$

$$v_2(t) = 15 \cos\left(377t + \frac{\pi}{12}\right) = 15 \cos\frac{\pi}{12} \cos(377t) - 15 \sin\frac{\pi}{12} \sin(377t) \text{ V}$$

Combine like terms to obtain:

$$v_1(t) + v_2(t) = 15 \left( \cos\frac{\pi}{4} + \cos\frac{\pi}{12} \right) \cos(377t) - 15 \left( \sin\frac{\pi}{4} + \sin\frac{\pi}{12} \right) \sin(377t)$$

$$= 15[1.673 \cos(377t) - 0.966 \sin(377t)]$$

$$= 15\sqrt{(1.673)^2 + (0.966)^2} \times \cos\left[377t + \arctan\left(\frac{0.966}{1.673}\right)\right]$$

$$= 15 \left[ 1.932 \cos\left(377t + \frac{\pi}{6}\right) \right] = 28.98 \cos\left(377t + \frac{\pi}{6}\right) \text{ V}$$

The above expression is, of course, identical to the one obtained using phasor notation, but it required more computation. Phasor analysis often simplifies calculations.

### EXAMPLE 4.11 AC Superposition

#### Problem

Compute the voltages  $v_1(t)$  and  $v_2(t)$  in the circuit of Figure 4.36.

#### Solution

##### Known Quantities:

$$i_S(t) = 0.5 \cos[2\pi(100t)] \text{ A}$$

$$v_S(t) = 20 \cos[2\pi(1,000t)] \text{ V}$$

**Find:**  $v_1(t)$  and  $v_2(t)$ .

**Analysis:** Since the two sources are at different frequencies, we must compute a separate solution for each. Consider the current source first, with the voltage source set to zero (short circuit) as shown in Figure 4.37. The circuit thus obtained is a simple current divider. Write the source current in phasor notation:

$$\mathbf{I}_S(j\omega) = 0.5e^{j0} = 0.5\angle 0 \text{ A} \quad \omega = 2\pi 100 \text{ rad/s}$$

Then

$$\mathbf{V}_{R1}(\mathbf{I}_S) = \mathbf{I}_S \frac{R_2}{R_1 + R_2} R_1 = 0.5\angle 0 \left( \frac{50}{150 + 50} \right) 150 = 18.75\angle 0 \text{ V}$$

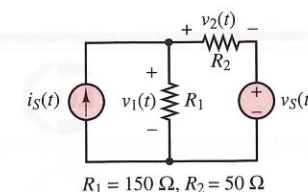
$$\omega = 2\pi(100) \text{ rad/s}$$

$$\mathbf{V}_{R2}(\mathbf{I}_S) = \mathbf{I}_S \frac{R_1}{R_1 + R_2} R_2 = 0.5\angle 0 \left( \frac{150}{150 + 50} \right) 50 = 18.75\angle 0 \text{ V}$$

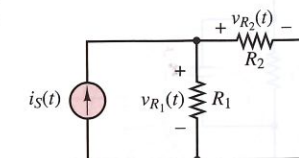
$$\omega = 2\pi(100) \text{ rad/s}$$

Next, consider the voltage source, with the current source set to zero (equivalent to an open circuit), as shown in Figure 4.38. First, write the source voltage in phasor notation:

$$\mathbf{V}_S(j\omega) = 20e^{j0} = 20\angle 0 \text{ V} \quad \omega = 2\pi(1,000) \text{ rad/s}$$



**Figure 4.36**



**Figure 4.37**

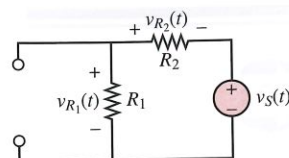


Figure 4.38

Then, apply the voltage divider law, to obtain:

$$\mathbf{V}_{R1}(\mathbf{V}_S) = \mathbf{V}_S \frac{R_1}{R_1 + R_2} = 20 \angle 0 \left( \frac{150}{150 + 50} \right) = 15 \angle 0 \text{ V}$$

$$\omega = 2\pi(1,000) \text{ rad/s}$$

$$\mathbf{V}_{R2}(\mathbf{V}_S) = -\mathbf{V}_S \frac{R_2}{R_1 + R_2} = -20 \angle 0 \left( \frac{50}{150 + 50} \right) = -5 \angle 0 = 5 \angle \pi \text{ V}$$

$$\omega = 2\pi(1,000) \text{ rad/s}$$

The voltage across each resistor is obtained by adding the contributions from each source and converting the equivalent phasor to the time domain:

$$\mathbf{V}_{R1} = \mathbf{V}_{R1}(\mathbf{I}_S) + \mathbf{V}_{R1}(\mathbf{V}_S)$$

$$v_1(t) = 18.75 \cos[2\pi(100t)] + 15 \cos[2\pi(1,000t)] \text{ V}$$

and

$$\mathbf{V}_{R2} = \mathbf{V}_{R2}(\mathbf{I}_S) + \mathbf{V}_{R2}(\mathbf{V}_S)$$

$$v_2(t) = 18.75 \cos[2\pi(100t)] + 5 \cos[2\pi(1,000t) + \pi] \text{ V}$$

**Comment:** It is impossible to further simplify the final expression because the two components are at different frequencies.



### EXAMPLE 4.12 Impedance of a Practical Capacitor

#### Problem

A practical capacitor can be modeled by an ideal capacitor in parallel with a resistor. The parallel resistance represents leakage losses in the capacitor that can be quite significant. Find the impedance of a practical capacitor at the radian frequency  $\omega = 377 \text{ rad/s}$  (60 Hz). How will the impedance change if the capacitor is used at a much higher frequency, say, 800 kHz?

#### Solution

**Known Quantities:** Figure 4.39;  $C_1 = 0.001 \mu\text{F} = 1 \times 10^{-9} \text{ F}$ ;  $R_1 = 1 \text{ M}\Omega$ .

**Find:** The equivalent impedance  $\mathbf{Z}_1$  across the parallel elements.

**Analysis:** Combine the two impedances in parallel to determine the equivalent impedance.

$$\mathbf{Z}_1 = R_1 \parallel \frac{1}{j\omega C_1} = \frac{R_1(1/j\omega C_1)}{R_1 + 1/j\omega C_1} = \frac{R_1}{1 + j\omega C_1 R_1}$$

Substitute numerical values to find:

$$\begin{aligned} \mathbf{Z}_1(\omega = 377) &= \frac{10^6}{1 + j377 \times 10^6 \times 10^{-9}} = \frac{10^6}{1 + j0.377} \\ &= 9.3571 \times 10^5 \angle (-0.3605) \Omega \end{aligned}$$

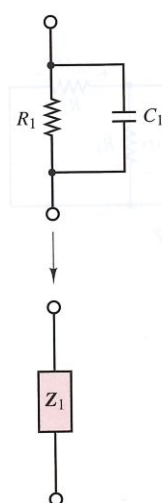


Figure 4.39

The impedance of the capacitor alone at this frequency is:

$$\mathbf{Z}_{C1}(\omega = 377) = \frac{1}{j377 \times 10^{-9}} = 2.6525 \times 10^6 \angle (-1.5708) \Omega$$

When the frequency is increased to 800 kHz, or  $1600\pi \times 10^3 \text{ rad/s}$ —a radio frequency in the AM range—the impedance changes to:

$$\begin{aligned} \mathbf{Z}_1(\omega = 1600\pi \times 10^3) &= \frac{10^6}{1 + j1600\pi \times 10^3 \times 10^{-9} \times 10^6} \\ &= \frac{10^6}{1 + j1600\pi} = 198.9 \angle (-1.5706) \Omega \end{aligned}$$

The impedance of the capacitor alone at this frequency would be

$$\mathbf{Z}_{C1}(\omega = 1600\pi \times 10^3) = \frac{1}{j1600\pi \times 10^3 \times 10^{-9}} = 198.9 \angle (-1.5708) \Omega$$

Now, the impedances  $\mathbf{Z}_1$  and  $\mathbf{Z}_{C1}$  are virtually identical. Thus, the effect of the parallel resistance is negligible at high frequencies.

**Comments:** For elements in parallel, the element with the smallest impedance tends to dominate the equivalent impedance across two nodes. At the lower frequency (corresponding to the well-known 60-Hz AC power frequency) the impedance of the resistor is roughly 38 percent smaller than that of the ideal capacitor. Thus, the resistor tends to dominate the equivalent impedance; in fact, the equivalent impedance is only 6.5 percent smaller than the resistance and so the practical and ideal capacitors are substantially different. At the higher frequency, the impedance of the ideal capacitor is much smaller than the resistance, which effectively acts as an open-circuit. The equivalent impedance is dominated by the ideal capacitor. At frequencies above and below  $\omega = 1/RC$ , the network is capacitive and resistive, respectively. This example suggests that the behavior of a network may depend heavily on frequency.

### EXAMPLE 4.13 Impedance of a Practical Inductor

#### Problem

Figure 4.40 shows a toroidal (doughnut-shaped) inductor. A practical inductor can be modeled as an ideal inductor in series with a resistor, as shown in Figure 4.41. The series resistance represents the resistance of the wire. Find the range of frequencies over which the impedance of the practical inductor is largely inductive (i.e., due to the inductance in the circuit). Consider the impedance to be inductive if it is at least 10 times larger than the resistance.

#### Solution

**Known Quantities:**  $L = 0.098 \text{ H}$ ; lead length =  $2 \times 10 \text{ cm}$ ;  $n = 250$  turns; wire is 30-gauge. Resistance of 30-gauge wire =  $0.344 \Omega/\text{m}$ .

**Find:** The range of frequencies over which the practical inductor acts nearly as an ideal inductor.

**Analysis:** To determine the equivalent resistance of the practical inductor, use the cross section of the toroid to estimate the length  $l_w$  of the wire coil:

$$l_w = 250(2 \times 0.25 + 2 \times 0.5) = 375 \text{ cm}$$

$$\text{Total length} = 375 + 20 = 395 \text{ cm}$$

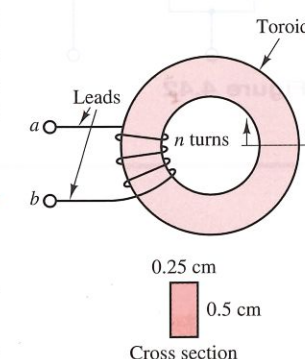


Figure 4.40 A practical inductor

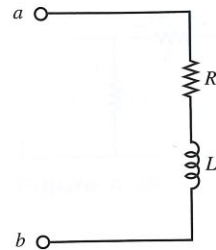


Figure 4.41

Thus, the total resistance is:

$$R = 0.344 \, \Omega/\text{m} \times 3.95 \, \text{m} = 1.36 \, \Omega$$

To determine the range of frequencies,  $\omega$ , over which the impedance  $j\omega L$  of the ideal inductor is  $10\times$  greater than  $1.36 \, \Omega$ :

$$\omega L > 13.6 \quad \text{or} \quad \omega > \frac{13.6}{L} = \frac{13.6}{0.098} = 139 \, \text{rad/s}$$

In terms of cyclical frequency, the range is  $f = \omega/2\pi > 22 \, \text{Hz}$ .

**Comments:** For elements in series, the element with the largest impedance tends to dominate the equivalent impedance across two nodes. At frequencies above  $139 \, \text{rad/s}$  the impedance of the inductor is at least  $10\times$  greater than the resistance and the resistance is insignificant. (Remember the 10:1 rule.) At lower frequencies, the resistance is significant; at very low frequencies ( $\omega L \ll R$ ), the impedance of the inductor effectively acts as a short-circuit and is negligible. At high frequencies, the separation between the insulated coil wires begins to exhibit significant capacitance and so the model should be modified accordingly.



#### EXAMPLE 4.14 Impedance of a Series-Parallel Network

##### Problem

Find the equivalent impedance of the circuit shown in Figure 4.42.

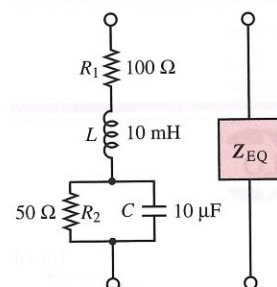


Figure 4.42

##### Solution

**Known Quantities:**  $\omega = 10^4 \, \text{rad/s}$ ;  $R_1 = 100 \, \Omega$ ;  $L = 10 \, \text{mH}$ ;  $R_2 = 50 \, \Omega$ ;  $C = 10 \, \mu\text{F}$ .

**Find:** The equivalent impedance of the series-parallel circuit.

**Analysis:** The equivalent impedance  $Z_{||}$  of  $R_2$  in parallel with  $C$  is:

$$\begin{aligned} Z_{||} &= R_2 \parallel \frac{1}{j\omega C} = \frac{R_2(1/j\omega C)}{R_2 + 1/j\omega C} = \frac{R_2}{1 + j\omega C R_2} \\ &= \frac{50}{1 + j10^4 \times 10 \times 10^{-6} \times 50} = \frac{50}{1 + j5} = 1.92 - j9.62 \\ &= 9.81 \angle (-1.3734) \, \Omega \end{aligned}$$

To determine the equivalent impedance  $Z_{eq}$  across the entire network:

$$\begin{aligned} Z_{eq} &= R_1 + j\omega L + Z_{||} = 100 + j10^4 \times 10^{-2} + 1.92 - j9.62 \\ &= 101.92 + j90.38 = 136.2 \angle 0.725 \, \Omega \end{aligned}$$

**Comment:** At  $\omega = 10^4 \, \text{rad/s}$ , the impedance across the network is inductive since the reactance is positive (or, equivalently, the phase angle is positive). (See Figure 4.31.)

#### EXAMPLE 4.15 Admittance

##### Problem

Find the equivalent admittance across each of the two networks shown in Figure 4.43.

##### Solution

**Known Quantities:**  $\omega = 2\pi \times 10^3 \, \text{rad/s}$ ;  $R_1 = 50 \, \Omega$ ;  $L = 16 \, \text{mH}$ ;  $R_2 = 100 \, \Omega$ ;  $C = 3 \, \mu\text{F}$ .

**Find:** The equivalent admittance across each of the two networks.

**Analysis:** Network (a): First, determine the equivalent impedance across the network  $ab$ :

$$Z_{ab} = R_1 + j\omega L$$

To obtain the admittance, compute the inverse of  $Z_{ab}$  by multiplying the numerator and denominator by the complex conjugate of the denominator:

$$Y_{ab} = \frac{1}{Z_{ab}} = \frac{1}{R_1 + j\omega L} = \frac{R_1 - j\omega L}{R_1^2 + (\omega L)^2}$$

Substitute numerical values to find:

$$\begin{aligned} Y_{ab} &= \frac{1}{50 + j2\pi \times 10^3 \times 0.016} = \frac{50 - j(2\pi \times 10^3)(0.016)}{50^2 + (2\pi \times 10^3)^2(0.016)^2} \\ &= 3.966 \times 10^{-3} - j7.975 \times 10^{-3} \, \text{S} \end{aligned}$$

Network (b): First, determine the equivalent impedance across the network  $ab$ :

$$Z_{ab} = R_2 \parallel \frac{1}{j\omega C} = \frac{R_2(1/j\omega C)}{R_2 + 1/j\omega C}$$

Multiply the numerator and denominator by  $j\omega C$  to find:

$$Z_{ab} = \frac{R_2}{1 + j\omega R_2 C}$$

The inverse of  $Z_{ab}$  is the admittance:

$$Y_{ab} = \frac{1}{Z_{ab}} = \frac{1 + j\omega R_2 C}{R_2} = \frac{1}{R_2} + j\omega C = 0.01 + j0.019 \, \text{S}$$

**Comment:** The units of admittance, siemens (S), are the same as the units of conductance.

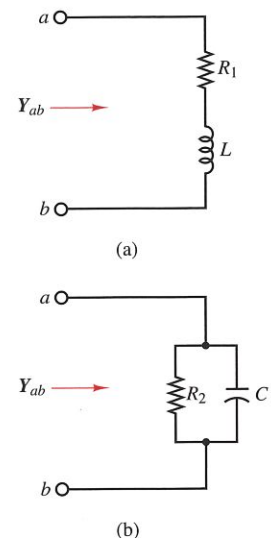


Figure 4.43

#### CHECK YOUR UNDERSTANDING

Add the sinusoidal voltages  $v_1(t) = A \cos(\omega t + \phi)$  and  $v_2(t) = B \cos(\omega t + \theta)$  using phasor notation, and then convert back to time-domain form.

- $A = 1.5 \, \text{V}$ ,  $\phi = 10^\circ$ ;  $B = 3.2 \, \text{V}$ ,  $\theta = 25^\circ$ .
- $A = 50 \, \text{V}$ ,  $\phi = -60^\circ$ ;  $B = 24 \, \text{V}$ ,  $\theta = 15^\circ$ .

Answers: (a)  $v_1 + v_2 = 4.67 \cos(\omega t + 0.353 \, \text{rad})$   
(b)  $v_1 + v_2 = 60.8 \cos(\omega t - 0.656 \, \text{rad})$

## CHECK YOUR UNDERSTANDING

Add the sinusoidal currents  $i_1(t) = A \cos(\omega t + \phi)$  and  $i_2(t) = B \cos(\omega t + \theta)$  for

- $A = 0.09 \text{ A}$ ,  $\phi = 72^\circ$ ;  $B = 0.12 \text{ A}$ ,  $\theta = 20^\circ$ .
- $A = 0.82 \text{ A}$ ,  $\phi = -30^\circ$ ;  $B = 0.5 \text{ A}$ ,  $\theta = -36^\circ$ .

Answers: (a)  $i_1 + i_2 = 0.19 \cos(\omega t + 0.733)$ ; (b)  $i_1 + i_2 = 1.32 \cos(\omega t - 0.5633)$

## CHECK YOUR UNDERSTANDING

Compute the equivalent impedance across the network of Example 4.14 for  $\omega = 1,000$  and  $100,000 \text{ rad/s}$ .

Find the reactance across the parallel  $R_2C$  network of Example 4.14 at the frequency  $\omega = 10 \text{ rad/s}$  and calculate its equivalent capacitance.

Answers:  $Z(1,000) = 140 - j10$ ;  $Z(100,000) = 100 + j999$ ;  $X_{||} = 0.25$ ;  $C = 0.4 \text{ F}$

## CHECK YOUR UNDERSTANDING

Compute the equivalent admittance across the network of Example 4.14.

Answer:  $Y_{eq} = 5.492 \times 10^{-3} - j4.871 \times 10^{-3}$

## 4.6 AC CIRCUIT ANALYSIS

Phasors and the concept of impedance, in particular, facilitate the solution of AC circuits by making it possible to use the same solution methods developed in Chapter 3 for DC circuits. Those methods are explored for AC circuits containing linear passive circuit elements ( $R$ ,  $L$ ,  $C$ ) excited by a sinusoidal source. Figure 4.44 depicts one such circuit, represented in both conventional time-domain and phasor-impedance forms.

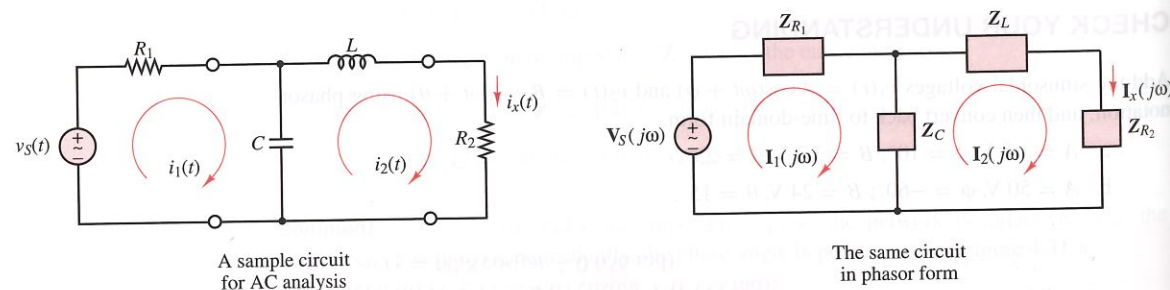


Figure 4.44 An AC circuit

The first step in AC circuit analysis is to convert all sources to phasor form and use the frequency of excitation to determine the impedance of each circuit element. Each impedance element will have an amplitude and a phase, both of which may depend upon the excitation frequency  $\omega$ .

The second step is to apply the same solution methods previously explored in Chapters 2 and 3, treating each impedance element in the same manner that resistors were treated in those chapters. The only difference is that AC circuit phasor analysis involves complex arithmetic whereas DC circuit analysis involves scalar arithmetic. For example, voltage division for two impedance elements in series will have the form:

$$\frac{V_1}{V_2} = \frac{Z_1}{Z_2}$$

Similarly, current division for two impedance elements in parallel will have the form:

$$\frac{I_1}{I_2} = \frac{Z_2}{Z_1}$$

Compare these expressions to those found in Chapter 2 for resistors in series and parallel. The applications of KVL, KCL, Ohm's law, node and mesh analyses, Thévenin's theorem, Norton's theorem, superposition, and source transformations to AC circuits are all identical to their applications in DC circuits, except that impedance  $Z$  and source phasors take the place of resistance and source scalars.

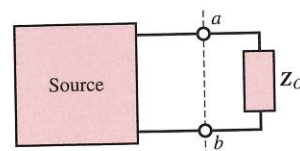
The solution of an AC circuit problem will be a phasor, in general. Thus, the third and last step is to convert the solution to its time-domain form. In effect, the use of phasor notation is but an intermediate step that facilitates the computation of the final answer.

It is worth noting that although it is possible to extend the methods of node and mesh analysis to AC circuits, the resulting simultaneous complex equations will usually be difficult to solve without the aid of a scientific calculator or computer, even for relatively simple circuits. In addition, these methods lend relatively little insight into the nature of the circuit. On the other hand, it is very useful to extend the concept of equivalent networks to the AC case and to make use of complex Thévenin and Norton equivalent impedances.

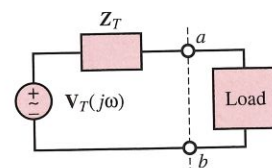
## FOCUS ON PROBLEM SOLVING

## AC CIRCUIT ANALYSIS

1. Identify the sinusoidal sources in a circuit and note the excitation frequencies.
2. Convert the sources to phasor form.
3. Use the excitation frequency to determine the impedance of each circuit element.
4. Solve the resulting phasor circuit, using an appropriate solution method, such as Thévenin's theorem, Norton's theorem, superposition, source transformation, nodal analysis, or mesh analysis. Take care to execute any complex arithmetic properly. Represent the solution as a phasor.
5. Convert the phasor solution to its time-domain form.



(a) Source-load perspective



(b) Simplified circuit

**Figure 4.45** AC circuit simplification using Thévenin's theorem

### AC Equivalent Circuits

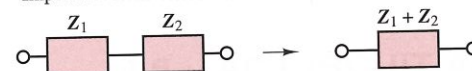
The concept of an equivalent circuit is equally useful in AC and DC circuit analyses. Figure 4.45(a) depicts the source-load perspective first introduced in Chapter 2. In the figure, the overall circuit is divided into two parts: a load and a source. Typically, the load is the element or circuit segment of interest to the analyst. The source is everything else not included in the load. The source and load are connected at two terminals  $a$  and  $b$ .

Thévenin's or Norton's theorem can be used to simplify the source network as shown in Figure 4.45(b). Notice that the Thévenin equivalent source is composed of two phasors: an independent voltage source  $V_T(j\omega)$  in series with an equivalent impedance  $Z_T(j\omega)$ . The voltage  $V_o$  across the load  $Z_o(j\omega)$  can be found by voltage division.

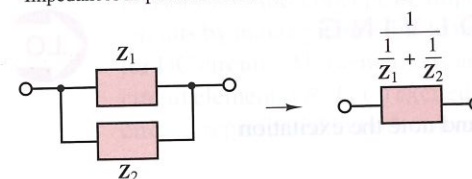
$$\frac{V_o}{V_S} = \frac{Z_o}{Z_o + Z_S}$$

Notice that the approach and form of this solution is exactly the same as that previously presented in Chapter 3 for DC circuits. Again, the only difference is the use of impedance instead of resistance. The independent Thévenin voltage source  $V_T$  is the open-circuit voltage  $V_{OC}$  across the terminals  $a$  and  $b$  of the source network. The Thévenin equivalent impedance  $Z_T$  is found by setting to zero all independent voltage and current sources in the source network and then finding the equivalent impedance  $Z_{ab}$  between the source network terminals. Once again, a zero voltage source is a short-circuit and a zero current source is an open-circuit. The rules for finding the equivalent impedance and admittance of series and parallel networks are shown in Figure 4.46.

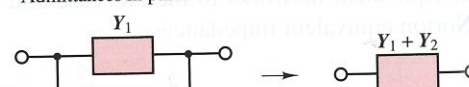
Impedances in series add:



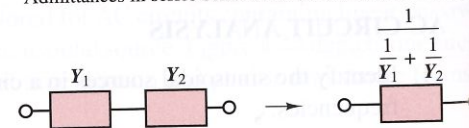
Impedances in parallel behave like resistors in parallel:



Admittances in parallel add:

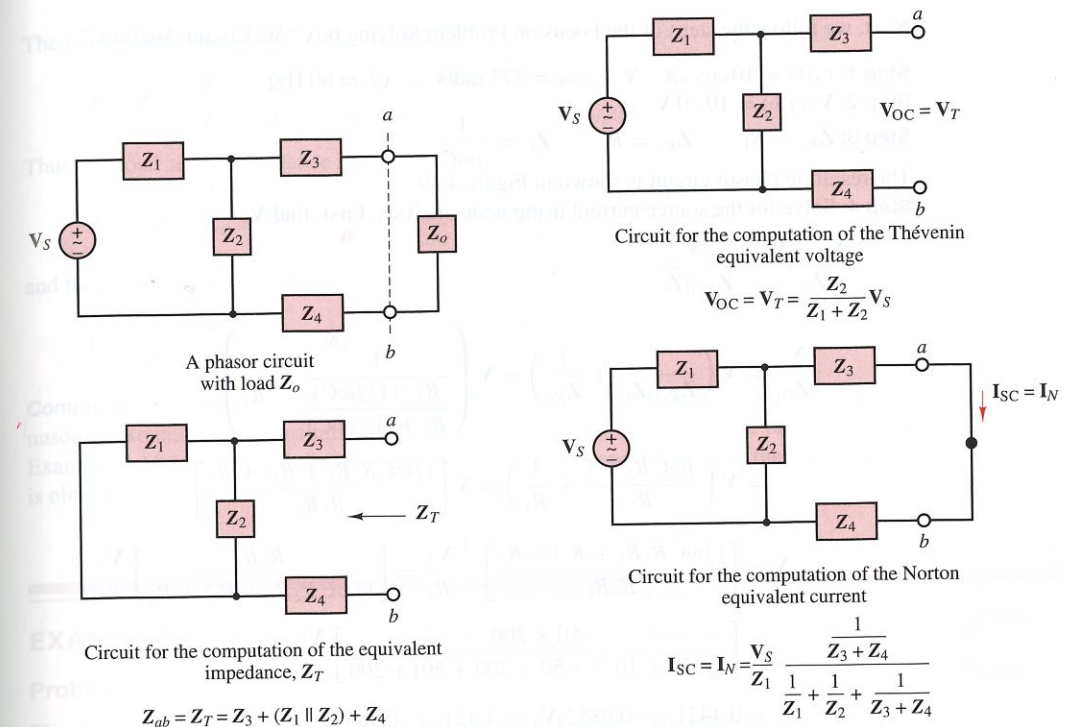


Admittances in series behave like conductances in series:



**Figure 4.46** Rules for impedance and admittance reduction

The Norton equivalent source is also composed of two phasors: an independent current source  $I_N$  in parallel with the same equivalent impedance  $Z_N(j\omega) = Z_T(j\omega)$ . Figure 4.47 depicts a somewhat complicated AC circuit partitioned into source and load networks, and how  $V_T$ ,  $I_N$ , and  $Z_T$  are found. The example problems that follow clarify some of the finer points in the calculation of such equivalent circuits. The details of the complex arithmetic are also explored.



**Figure 4.47** Reduction of AC circuit to equivalent form

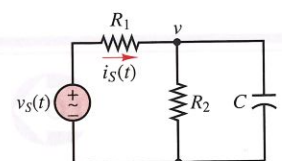
### EXAMPLE 4.16 Phasor Analysis of an AC Circuit

#### Problem

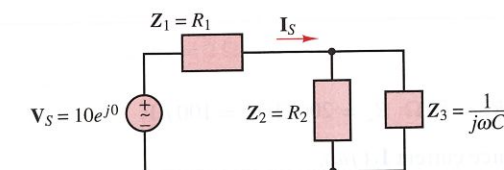
Apply the phasor analysis method to the circuit of Figure 4.48 to determine the source current.

#### Solution

**Known Quantities:** Figures 4.48, 4.49,  $v_S(t) = 10 \cos \omega t$ ;  $\omega = 377 \text{ rad/s}$ ;  $R_1 = 50 \Omega$ ;  $R_2 = 200 \Omega$ ;  $C = 100 \mu\text{F}$ .



**Figure 4.48**



**Figure 4.49**

**Find:** The source current  $i_S(t)$ .

**Analysis:** Define the voltage  $v$  at the top node, and use node analysis to determine  $v$ . Then observe that

$$i_S(t) = \frac{v_S(t) - v(t)}{R_1}$$

Next, we follow the steps of the Focus on Problem Solving box “AC Circuit Analysis.”

**Step 1:**  $v_S(t) = 10 \cos \omega t$  V  $\omega = 377$  rad/s ( $f = 60$  Hz)

**Step 2:**  $V_S(j\omega) = 10\angle 0$  V

**Step 3:**  $Z_{R1} = R_1$   $Z_{R2} = R_2$   $Z_C = \frac{1}{j\omega C}$

The resulting phasor circuit is shown in Figure 4.49.

**Step 4:** Solve for the source current using node analysis. First, find  $V$ :

$$\frac{V_S - V}{Z_{R1}} = \frac{V}{Z_{R2} \parallel Z_C}$$

$$\frac{V_S}{Z_{R1}} = V \left( \frac{1}{Z_{R2} \parallel Z_C} + \frac{1}{Z_{R1}} \right) = V \left( \frac{1}{\frac{R_2 \cdot (1/j\omega C)}{R_2 + (1/j\omega C)}} + \frac{1}{R_1} \right)$$

$$= V \left( \frac{j\omega C R_2 + 1}{R_2} + \frac{1}{R_1} \right) = V \left[ \frac{(j\omega C R_2 R_1 + R_1) + R_2}{R_1 R_2} \right]$$

$$V = \left[ \frac{(j\omega C R_2 R_1 + R_1) + R_2}{R_1 R_2} \right]^{-1} \frac{V_S}{R_1} = \left[ \frac{R_1 R_2}{(j\omega C R_2 R_1 + R_1) + R_2} \right] \frac{V_S}{R_1}$$

$$= \left[ \frac{50 \times 200}{(j377 \times 10^{-4} \times 50 \times 200 + 50) + 200} \right] \frac{V_S}{50}$$

$$= 0.4421\angle(-0.9852)V_S = 4.421\angle(-0.9852)$$

Then, compute  $I_S$ :

$$I_S = \frac{V_S - V}{Z_{R1}} = \frac{10\angle 0 - 4.421\angle(-0.9852)}{50} = 0.1681\angle(0.4537)$$

**Step 5:** Finally, convert the solution to its time-domain form:

$$i_S(t) = 0.1681 \cos(377t + 0.4537)$$



### EXAMPLE 4.17 An AC Circuit with an Arbitrary Sinusoidal Input

#### Problem

Determine the general solution of Example 4.16 for any sinusoidal source,  $A \cos(\omega t + \phi)$ .

#### Solution

**Known Quantities:**  $R_1 = 50 \Omega$ ;  $R_2 = 200 \Omega$ ,  $C = 100 \mu\text{F}$ .

**Find:** The phasor source current  $I_S(j\omega)$ .

**Analysis:** Since the radian frequency is arbitrary, it will be impossible to determine a numerical answer. The answer will be a function of  $\omega$ . The source phasor is:  $V_S(j\omega) = A\angle\phi$ . The impedances will be  $Z_{R1} = 50 \Omega$ ;  $Z_{R2} = 200 \Omega$ ;  $Z_C = -j10^4/\omega \Omega$ . Note that the impedance of the capacitor is a function of  $\omega$ .

Observe that the source current is given by:

$$I_S = \frac{V_S}{Z_{R1} + Z_{R2} \parallel Z_C}$$

The parallel impedance  $Z_{R2} \parallel Z_C$  is given by:

$$Z_{R2} \parallel Z_C = \frac{Z_{R2} \times Z_C}{Z_{R2} + Z_C} = \frac{200 \times 10^4/j\omega}{200 + 10^4/j\omega} = \frac{2 \times 10^6}{10^4 + j\omega 200} \Omega$$

Thus, the total series impedance is:

$$Z_{R1} + Z_{R2} \parallel Z_C = 50 + \frac{2 \times 10^6}{10^4 + j\omega 200} = \frac{2.5 \times 10^6 + j\omega 10^4}{10^4 + j\omega 200} \Omega$$

and the phasor source current is:

$$I_S = \frac{V_S}{Z_{R1} + Z_{R2} \parallel Z_C} = A\angle\phi \frac{10^4 + j\omega 200}{2.5 \times 10^6 + j\omega 10^4} \text{ A}$$

**Comment:** The expression obtained in this example can be evaluated for an arbitrary sinusoidal excitation, by substituting numerical values for  $A$ ,  $\phi$ , and  $\omega$ . Use the values from Example 4.16 ( $A = 10$  V,  $\phi = 0$  rad,  $\omega = 377$  rad/s) and verify that the same answer is obtained.

### EXAMPLE 4.18 Solving an AC Circuit by Node Analysis



#### Problem

The electrical characteristics of electric motors (which are described in greater detail in the last two chapters of this book) can be approximated by a series  $RL$  circuit. In this problem, a voltage source provides current to two different motors (Figure 4.50).

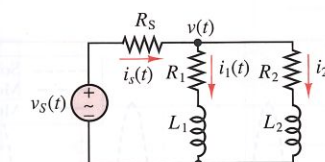


Figure 4.50 AC circuit used to demonstrate node analysis

#### Solution

**Known Quantities:**  $R_S = 0.5 \Omega$ ;  $R_1 = 2 \Omega$ ;  $R_2 = 0.2 \Omega$ ;  $L_1 = 0.1$  H;  $L_2 = 20$  mH;  $v_S(t) = 155 \cos(377t)$  V.

**Find:** The motor load currents  $i_1(t)$  and  $i_2(t)$ .

**Analysis:** First, calculate the impedances of the source and of each motor:

$$Z_S = 0.5 \Omega$$

$$Z_1 = 2 + j377 \times 0.1 = 2 + j37.7 = 37.75\angle 1.52 \Omega$$

$$Z_2 = 0.2 + j377 \times 0.02 = 0.2 + j7.54 = 7.54\angle 1.54 \Omega$$

The source voltage is  $V_S = 155\angle 0$  V.

Next, apply KCL at the top right node, with the aim of solving for the node voltage  $V$ :

$$\begin{aligned}\frac{V_S - V}{Z_S} &= \frac{V}{Z_1} + \frac{V}{Z_2} \\ \frac{V_S}{Z_S} &= \frac{V}{Z_S} + \frac{V}{Z_1} + \frac{V}{Z_2} = V \left( \frac{1}{Z_S} + \frac{1}{Z_1} + \frac{1}{Z_2} \right) \\ V &= \left( \frac{1}{Z_S} + \frac{1}{Z_1} + \frac{1}{Z_2} \right)^{-1} \frac{V_S}{Z_S} = \left( \frac{1}{0.5} + \frac{1}{2 + j37.7} + \frac{1}{0.2 + j7.54} \right)^{-1} \frac{V_S}{0.5} \\ &= 154.1 \angle 0.079^\circ \text{ V}\end{aligned}$$

It is now easy to find the phasor motor currents,  $I_1$  and  $I_2$ , from the phasor node voltage  $V$ .

$$\begin{aligned}I_1 &= \frac{V}{Z_1} = \frac{154 \angle 0.079^\circ}{2 + j37.7} = 4.083 \angle -1.439^\circ \\ I_2 &= \frac{V}{Z_2} = \frac{154 \angle 0.079^\circ}{0.2 + j7.54} = 20.44 \angle -1.465^\circ\end{aligned}$$

Finally, write the time-domain expressions for the currents:

$$\begin{aligned}i_1(t) &= 4.083 \cos(377t - 1.439^\circ) \text{ A} \\ i_2(t) &= 20.44 \cos(377t - 1.465^\circ) \text{ A}\end{aligned}$$

Figure 4.51 depicts the source voltage (scaled down by a factor of 10) and the two motor currents.

**Comment:** Note the phase shift between the source voltage and the two motor currents.

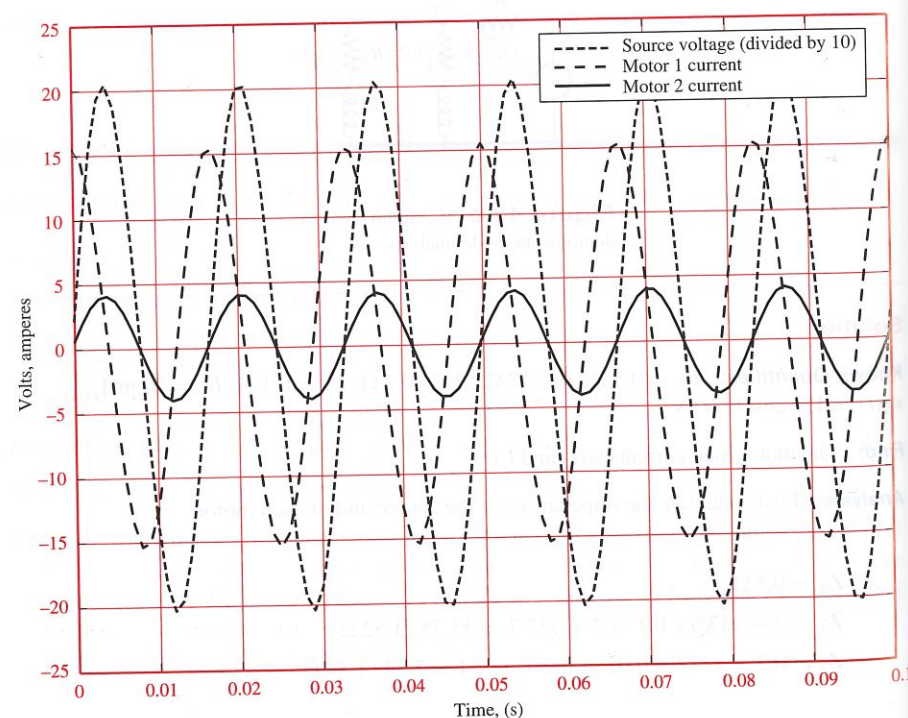


Figure 4.51 Source voltage and motor currents for Example 4.18

### EXAMPLE 4.19 Using Thévenin's Theorem to Solve an AC Circuit

#### Problem

Compute the Thévenin equivalent of the circuit of Figure 4.52.

#### Solution

**Known Quantities:**  $Z_1 = 5 \Omega$ ;  $Z_2 = j20 \Omega$ ,  $v_S(t) = 110 \cos(377t)$  V.

**Find:** Thévenin equivalent circuit.

**Analysis:** Find the equivalent impedance seen by the load  $Z_O$ . Remove the load, replace the independent voltage source with a short-circuit, and compute the equivalent impedance between terminals  $a$  and  $b$  shown in Figure 4.53.

$$Z_T = Z_1 || Z_2 = \frac{Z_1 \times Z_2}{Z_1 + Z_2} = \frac{5 \times j20}{5 + j20} = 4.71 + j1.176 \Omega$$

Next, find the open-circuit voltage between terminals  $a$  and  $b$ :

$$V_T = \frac{Z_2}{Z_1 + Z_2} V_S = \frac{j20}{5 + j20} 110 \angle 0^\circ = \frac{20 \angle \pi/2}{20.6 \angle 1.326} 110 \angle 0^\circ = 106.7 \angle 0.245^\circ \text{ V}$$

The complete simplified circuit is shown in Figure 4.54.

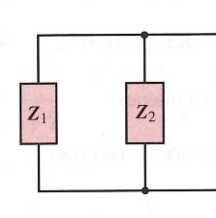


Figure 4.53

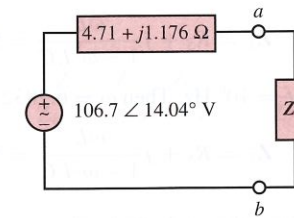


Figure 4.54

**Comments:** Note that the procedure used to simplify the circuit is identical to that used for resistive circuits, the only difference being in the use of complex impedances in place of resistances.

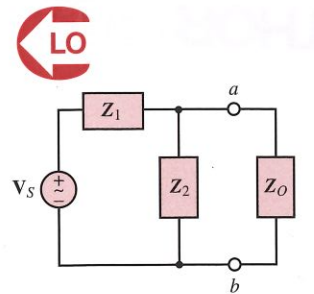
### EXAMPLE 4.20 Thévenin's Theorem Applied to an AC Circuit

#### Problem

Determine the Thévenin equivalent circuit seen by the load in the circuit of Figure 4.55 when the input sinusoidal voltage is (a) at a frequency of  $10^3$  Hz and (b) at a frequency of  $10^6$  Hz.

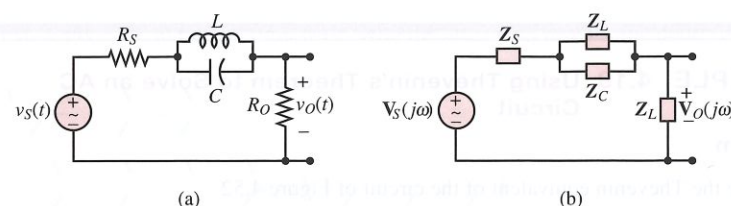
#### Solution

**Known Quantities:** The values of the resistances  $R_S = R_O = 50 \Omega$ , capacitance  $C = 0.1 \mu\text{F}$ , inductance  $L = 10 \text{ mH}$ .



$$V_S = 110 \angle 0^\circ \text{ V}, \quad Z_1 = 5 \Omega, \quad Z_2 = j20 \Omega$$

Figure 4.52



**Figure 4.55** (a) Circuit for Example 4.20; (b) same circuit ready for phasor analysis

**Analysis:** First, convert the circuit to phasor form, as shown in Figure 4.55(b). Next, compute the Thévenin equivalent impedance with the load removed:

$$\begin{aligned} Z_T &= Z_S + Z_L \parallel Z_C = R_S + \frac{j\omega L \times 1/j\omega C}{j\omega L + 1/j\omega C} \\ &= R_S + \frac{j\omega L}{j\omega L \times j\omega C + 1} = R_S + j \frac{\omega L}{1 - \omega^2 LC} \end{aligned}$$

Observe that the Thévenin equivalent voltage is equal to the source voltage, since once the load is removed, the current is zero in the resulting network and the voltage drop across the remaining impedances is zero. Thus:

$$V_T = V_S$$

Next, evaluate the Thévenin equivalent at each of the two frequencies.

a. Let  $f = 10^3$  Hz. Then  $\omega = 6.2832 \times 10^3$ . At this frequency,

$$Z_T = R_S + j \frac{\omega L}{1 - \omega^2 LC} = 50 + j65.414 = 82.33 \angle 0.9182$$

b. Let  $f = 10^6$  Hz. Then  $\omega = 6.2832 \times 10^6$ . At this frequency,

$$Z_T = R_S + j \frac{\omega L}{1 - \omega^2 LC} = 50 - j1.5916 = 50 \angle (-0.0318)$$

**Comments:** Note that at the higher frequency the equivalent impedance is very close to that of the resistor  $R_S$ . This result is due to the action of the capacitor and inductor, which behave like short- and open-circuits, respectively, at high frequency. The very small impedance of the capacitor at high frequency dominates the parallel equivalent impedance.



### EXAMPLE 4.21 Solution of an AC Circuit by Mesh Analysis

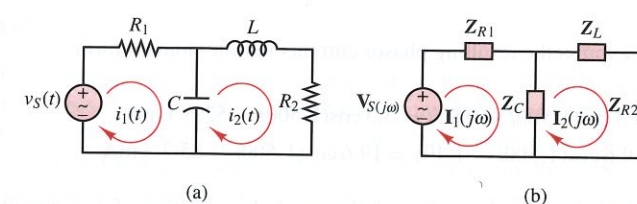
#### Problem

Determine the currents  $i_1(t)$  and  $i_2(t)$  in the circuit of Figure 4.56, using node analysis.

#### Solution

**Known Quantities:** The values of the circuit elements are  $R_1 = 100 \Omega$ ,  $R_2 = 75 \Omega$ ,  $C = 1 \mu\text{F}$ ,  $L = 0.5$  H. The value of the current source is  $v_S(t) = 15 \cos(1,500t)$  V.

**Analysis:** Follow the steps of the Focus on Problem Solving box “AC Circuit Analysis.”



**Figure 4.56** (a) Circuit for Example 4.21; (b) same circuit ready for phasor analysis

**Step 1:**  $v_S(t) = 15 \cos(1,500t)$  V  $\omega = 1,500$  rad/s

**Step 2:**  $V_S(j\omega) = 15 \angle 0$  V

**Step 3:**  $Z_{R1} = R_1$   $Z_{R2} = R_2$   $Z_C = \frac{1}{j\omega C}$   $Z_L = j\omega L$

The resulting phasor circuit is shown in Figure 4.56(b).

**Step 4:** The mesh equations are:

$$V_S(j\omega) - Z_{R1} I_1(j\omega) - Z_C [I_1(j\omega) - I_2(j\omega)] = 0 \quad \text{mesh 1}$$

$$Z_C [I_2(j\omega) - I_1(j\omega)] + Z_L I_2(j\omega) + Z_{R2} I_2(j\omega) = 0 \quad \text{mesh 2}$$

The matrix form of the equations is:

$$\begin{bmatrix} Z_{R1} + Z_C & -Z_C \\ -Z_C & Z_C + Z_L + Z_{R2} \end{bmatrix} \begin{bmatrix} I_1(j\omega) \\ I_2(j\omega) \end{bmatrix} = \begin{bmatrix} V_S(j\omega) \\ 0 \end{bmatrix}$$

Cramer's rule can be used to solve for the two currents:

$$I_1(j\omega) = \frac{\begin{vmatrix} V_S(j\omega) & -Z_C \\ 0 & Z_C + Z_L + Z_{R2} \end{vmatrix}}{\begin{vmatrix} Z_{R1} + Z_C & -Z_C \\ -Z_C & Z_C + Z_L + Z_{R2} \end{vmatrix}} = \frac{Z_C + Z_L + Z_{R2}}{(Z_{R1} + Z_C)(Z_C + Z_L + Z_{R2}) - Z_C^2} V_S(j\omega)$$

$$I_2(j\omega) = \frac{\begin{vmatrix} Z_{R1} + Z_C & V_S(j\omega) \\ -Z_C & 0 \end{vmatrix}}{\begin{vmatrix} Z_{R1} + Z_C & -Z_C \\ -Z_C & Z_C + Z_L + Z_{R2} \end{vmatrix}} = \frac{Z_C}{(Z_{R1} + Z_C)(Z_C + Z_L + Z_{R2}) - Z_C^2} V_S(j\omega)$$

**Step 5:** Substitute the impedance values in the previous expressions to find:

$$I_1(j\omega) = \frac{1/j\omega C + j\omega L + R_2}{(R_1 + 1/j\omega C)(1/j\omega C + j\omega L + R_2) - (1/j\omega C)^2} V_S(j\omega)$$

$$= \frac{j\omega C + (j\omega C)^2(j\omega L) + (j\omega C)^2 R_2}{(j\omega C R_1 + 1)[1 + (j\omega C)(j\omega L) + j\omega C R_2] - 1} V_S(j\omega)$$

$$I_2(j\omega) = \frac{1/j\omega C}{(R_1 + 1/j\omega C)(1/j\omega C + j\omega L + R_2) - (1/j\omega C)^2} V_S(j\omega)$$

$$= \frac{j\omega C}{(j\omega C R_1 + 1)[1 + (j\omega C)(j\omega L) + j\omega C R_2] - 1} V_S(j\omega)$$

Plug in numerical values to obtain:

$$I_1(j\omega) = 0.0033 \angle 0.92 = 0.0033 \angle 52.6^\circ \text{ A}$$

$$I_2(j\omega) = 0.0196 \angle -1.49 = 0.0196 \angle -85.3^\circ \text{ A}$$

**Step 6:** Finally, express the resulting phasor currents in time-domain form:

$$i_1(t) = 3.3 \cos(1,500t + 0.92) = 3.3 \cos(1,500t + 52.6^\circ) \text{ mA}$$

$$i_2(t) = 19.6 \cos(1,500t - 1.49) = 19.6 \cos(1,500t - 85.3^\circ) \text{ mA}$$

**Comments:** Note that the derivation of the symbolic equations for a circuit in phasor-impedance form using matrix techniques is no more involved than it would be for a resistive circuit. The only difference is the complex algebra manipulations.

### CHECK YOUR UNDERSTANDING

Compute the magnitude of the current  $\mathbf{I}_S(j\omega)$  of Example 4.17 if  $A = 10$  and  $\phi = 0$ , for  $\omega = 10, 10^2, 10^3, 10^4$ , and  $10^5$  rad/s. Are these results intuitively satisfying?

Answer:  $|\mathbf{I}_S| = 0.041 \text{ A}; 0.2 \text{ A}; 0.194 \text{ A}; 0.083 \text{ A}; 0.1 \text{ A}$ . As the frequency increases, the impedance of the capacitor decreases. In the limit  $\omega \rightarrow \infty$  the capacitor acts as a short-circuit. Thus, at sufficiently high frequency,  $|\mathbf{I}_S| \approx |\mathbf{V}_S|/R_1 = 0.2 \text{ A}$ .

### CHECK YOUR UNDERSTANDING

In the circuit of Example 4.18, assume that the branch  $R_2$  in series with  $L_2$  is the load. Find the Norton equivalent of the rest of the circuit.

Answer: Short-circuit current is  $310e^{j0} \text{ A}$ . Norton equivalent impedance is  $\approx 0.5e^{j0.013} \Omega$ .

### CHECK YOUR UNDERSTANDING

In Example 4.20, determine the value of the capacitor and inductor impedance at the two given frequencies. Compare these values to  $R_S$ . Do the results confirm what is stated in the Comments section?

Answer: At  $\omega = 2\pi \times 10^3 \text{ rad/s}$ ,  $\mathbf{Z}_L = j6.2832 \Omega$ ,  $\mathbf{Z}_C = -j1.5915 \Omega$ . At  $\omega = 2\pi \times 10^6 \text{ rad/s}$ ,  $\mathbf{Z}_L = j62.832 \Omega$ ,  $\mathbf{Z}_C = -j1.5915 \times 10^3 \Omega$ . At

### CHECK YOUR UNDERSTANDING

In Example 4.21, find the equivalent impedance of  $\mathbf{Z}_{R_1}$  in series with the equivalent impedance of  $\mathbf{Z}_C$  in parallel with  $\mathbf{Z}_L + \mathbf{Z}_{R_2}$  to calculate the voltage source current. Show that it is equal to the mesh current  $i_1(t)$ .

## Conclusion

This chapter introduced concepts and tools useful in the analysis of AC circuits. The importance of AC circuit analysis cannot be overemphasized, for a number of reasons. First, circuits made up of resistors, inductors, and capacitors constitute reasonable models for more complex devices, such as transformers, electric motors, and electronic amplifiers. Second, sinusoidal signals are ever-present in the analysis of many physical systems, not just circuits. Upon completion of this chapter a student will have learned to:

1. Compute currents, voltages, and energy stored in capacitors and inductors.
2. Calculate the average and root-mean-square value of an arbitrary (periodic) signal.
3. Write differential equations for circuits containing inductors and capacitors.
4. Convert time-domain sinusoidal voltages and currents to phasor notation, and vice versa, and represent circuits using impedances.
5. Apply the circuit analysis methods of Chapter 3 to AC circuits in phasor form.

## HOMEWORK PROBLEMS

### Section 4.1 Capacitors and Inductors

**4.1** The current through a 0.8-H inductor is given by  $i_L = \sin(100t + \frac{\pi}{4})$ . Write the expression for the voltage across the inductor.

**4.2** For each case shown below, derive the expression for the current through a 200- $\mu\text{F}$  capacitor.  $v_C(t)$  is the voltage across the capacitor.

- a.  $v_C(t) = 22 \cos(20t - \frac{\pi}{3}) \text{ V}$
- b.  $v_C(t) = -40 \cos(90t + \frac{\pi}{2}) \text{ V}$
- c.  $v_C(t) = 28 \cos(15t + \frac{\pi}{8}) \text{ V}$
- d.  $v_C(t) = 45 \cos(120t + \frac{\pi}{4}) \text{ V}$

**4.3** Derive the expression for the voltage across a 200-mH inductor when its current is:

- a.  $i_L = -2 \sin 10t \text{ A}$
- b.  $i_L = 2 \cos 3t \text{ A}$
- c.  $i_L = -10 \sin(50t - \frac{\pi}{4}) \text{ A}$
- d.  $i_L = 7 \cos(10t + \frac{\pi}{4}) \text{ A}$

**4.4** In the circuit shown in Figure P4.4, assume  $R = 1 \Omega$  and  $L = 2 \text{ H}$ . Also, let:

$$i(t) = \begin{cases} 0 & -\infty < t < 0 \\ t & 0 \leq t < 10 \text{ s} \\ 10 & 10 \leq t < \infty \end{cases}$$

Find the energy stored in the inductor for all time.

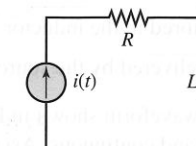


Figure P4.4

**4.5** Refer to Problem 4.4 and find the energy delivered by the source for all time.

**4.6** In the circuit shown in Figure P4.4, assume  $R = 2 \Omega$  and  $L = 4 \text{ H}$ . Also, let:

$$i(t) = \begin{cases} 0 & -\infty < t < 0 \\ 2t & 0 \leq t < 5 \text{ s} \\ 10 - 4t & 5 \leq t < 12 \text{ s} \\ 2 & 12 \leq t < \infty \end{cases}$$

Find:

- a. The energy stored in the inductor for all time.
- b. The energy delivered by the source for all time.

**4.7** In the circuit shown in Figure P4.7, assume  $R = 2 \Omega$  and  $C = 0.1 \text{ F}$ . Also, let:

$$v(t) = \begin{cases} 0 & \text{for } -\infty < t < 0 \\ t & \text{for } 0 \leq t < 10 \text{ s} \\ 10 & \text{for } 10 \leq t < \infty \end{cases}$$

Find the energy stored in the capacitor for all time.

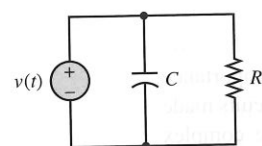


Figure P4.7

**4.8** Refer to Problem 4.7 and find the energy delivered by the source for all time.

**4.9** In the circuit shown in Figure P4.7, assume  $R = 4\ \Omega$  and  $C = 0.2\ \text{F}$ . Also, let:

$$v(t) = \begin{cases} 0 & -\infty < t < 0 \\ 4t & 0 \leq t < 4\ \text{s} \\ 2 - 0.5t & 4 \leq t < 10\ \text{s} \\ 0 & t > 10\ \text{s} \end{cases}$$

Find:

- The energy stored in the inductor for all time.
- The energy delivered by the source for all time.

**4.10** The voltage waveform shown in Figure P4.10 is piecewise linear and continuous. Assume the voltage is across a 20-mH inductor. Determine the current  $i_L(t)$  through the inductor, assuming  $i_L(0) = 50\ \text{mA}$ .

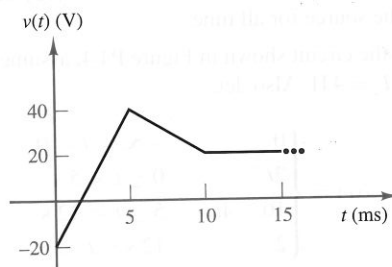


Figure P4.10

**4.11** The voltage waveform shown in Figure P4.10 is piecewise linear and continuous. Assume the voltage is across a 100- $\mu\text{F}$  capacitor. Determine the current  $i_C(t)$  through the capacitor even when the space between the capacitor plates is a perfect insulator. How is current through a capacitor different from leakage current?

**4.12** The voltage across a 0.5-mH inductor, plotted as a function of time, is shown in Figure P4.12. Determine the current through the inductor at  $t = 6\ \text{ms}$ .

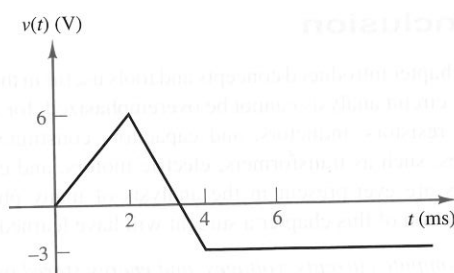


Figure P4.12

**4.13** Figure P4.13 shows the voltage across a capacitor plotted as a function of time where:

$$v_{PK} = 20\ \text{V} \quad T = 40\ \mu\text{s} \quad C = 680\ \text{nF}$$

Determine and plot the waveform for the current through the capacitor as a function of time. How is the current affected by the discontinuities in slope in the voltage waveform?

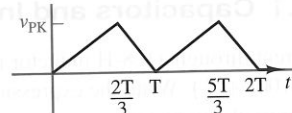


Figure P4.13

**4.14** The current through a 16- $\mu\text{H}$  inductor is zero at  $t = 0$ , and the voltage across the inductor (shown in Figure P4.14) is:

$$v(t) = \begin{cases} 0 & t \leq 0 \\ 3t^2 & 0 \leq t \leq 20\ \mu\text{s} \\ 1.2\ \text{nV} & t \geq 20\ \mu\text{s} \end{cases}$$

Determine the current through the inductor at  $t = 30\ \mu\text{s}$ .

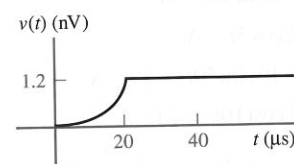


Figure P4.14

- 4.15** The voltage across a generic element X has the waveform shown in Figure P4.15. For  $0 < t < 10\ \text{ms}$ , determine and plot the current through X when it is a:
- 7- $\Omega$  resistor.
  - 0.5- $\mu\text{F}$  capacitor.
  - 7-mH inductor.

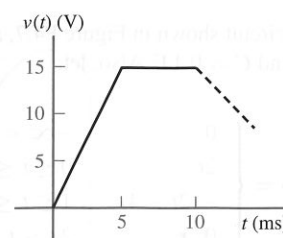


Figure P4.15

**4.16** The plots shown in Figure P4.16 are the voltage across and the current through an ideal capacitor. Determine its capacitance.

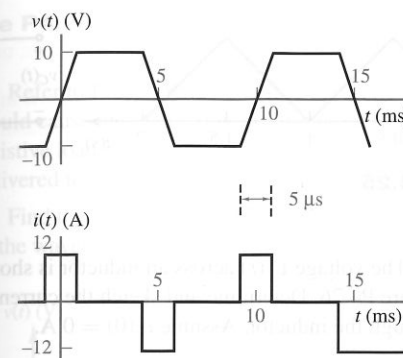


Figure P4.16

**4.17** The plots shown in Figure P4.17 are the voltage across and the current through an ideal inductor. Determine its inductance.

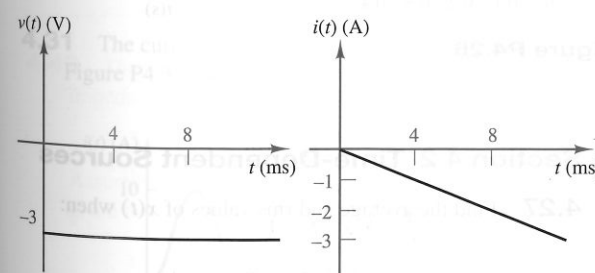


Figure P4.17

**4.18** The plots shown in Figure P4.18 are the voltage across and the current through an ideal capacitor. Determine its capacitance.

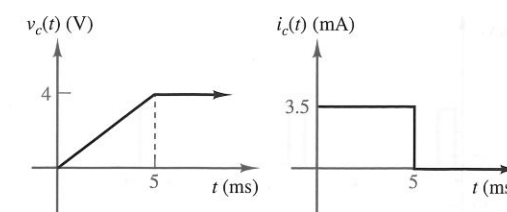


Figure P4.18

**4.19** The plots shown in Figure P4.19 are the voltage across and the current through an ideal capacitor. Determine its capacitance.

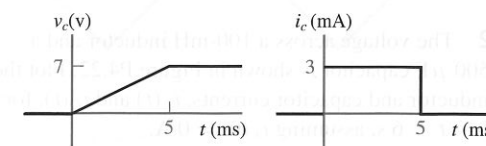


Figure P4.19

**4.20** The voltage  $v_L(t)$  across a 10-mH inductor is shown in Figure P4.20. Find the current  $i_L(t)$  through the inductor. Assume  $i_L(0) = 0\ \text{A}$ .

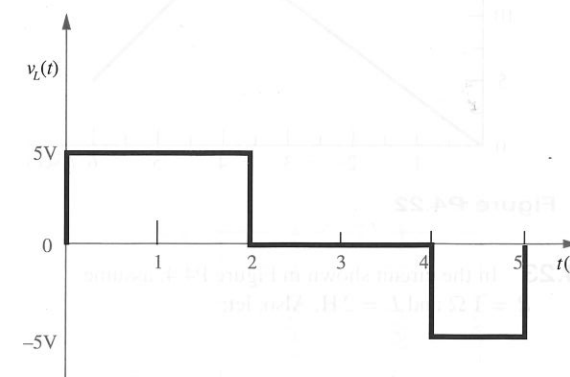


Figure P4.20

**4.21** The current through a 2-H inductor is plotted in Figure P4.21. Plot the inductor voltage  $v_L(t)$ .

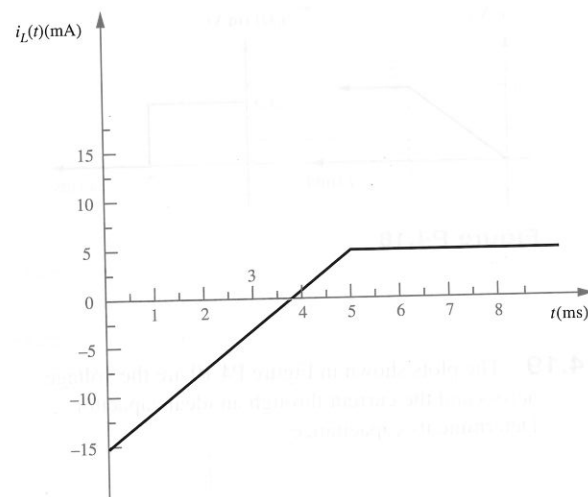


Figure P4.21

- 4.22** The voltage across a 100-mH inductor and a 500- $\mu$ F capacitor is shown in Figure P4.22. Plot the inductor and capacitor currents,  $i_L(t)$  and  $i_C(t)$ , for  $0 < t < 6$  s, assuming  $i_L(0) = 0$  A.

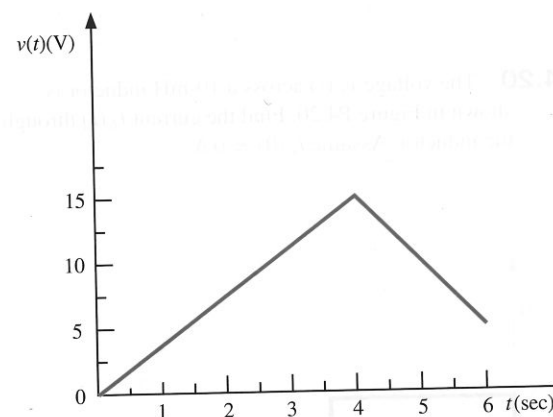


Figure P4.22

- 4.23** In the circuit shown in Figure P4.4, assume  $R = 1 \Omega$  and  $L = 2$  H. Also, let:

$$i(t) = \begin{cases} 0 & -\infty < t < 0 \\ t & 0 \leq t < 1 \text{ s} \\ -(t-2) & 1 \leq t < 2 \text{ s} \\ 0 & 2 \leq t < \infty \end{cases}$$

Find the energy stored in the inductor for all time.

- 4.24** In the circuit shown in Figure P4.7, assume  $R = 2 \Omega$  and  $C = 0.1$  F. Also, let:

$$v(t) = \begin{cases} 0 & -\infty < t \leq 0 \\ 2t & 0 \leq t \leq 1 \text{ s} \\ -(2t-4) & 1 \leq t \leq 2 \text{ s} \\ 0 & 2 \leq t < \infty \end{cases}$$

Find the energy stored in the capacitor for all time.

- 4.25** The voltage  $v_C(t)$  across a capacitor is shown in Figure P4.25. Determine and sketch the current  $i_C(t)$  through the capacitor.

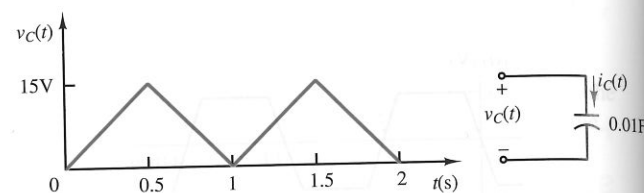


Figure P4.25

- 4.26** The voltage  $v_L(t)$  across an inductor is shown in Figure P4.26. Determine and sketch the current  $i_L(t)$  through the inductor. Assume  $i_L(0) = 0$  A.

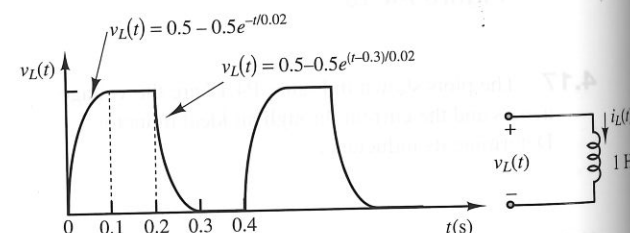


Figure P4.26

## Section 4.2 Time-Dependent Sources

- 4.27** Find the average and rms values of  $x(t)$  when:

$$x(t) = 3 \cos(7\omega t) + 4$$

- 4.28** The output voltage waveform of a controlled rectifier is shown in Figure P4.28. The input voltage waveform was a sinusoid of amplitude 110 V rms. Find the average and rms voltages of the output waveform in terms of the firing angle  $\theta$ .

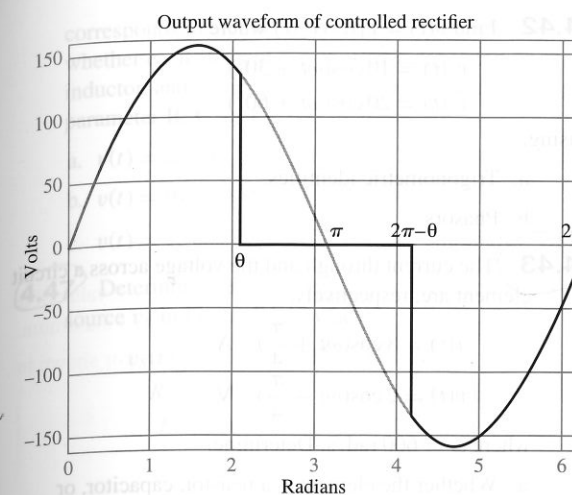


Figure P4.28

- 4.29** Refer to Problem 4.28 and find the angle  $\theta$  that would cause the rectified waveform to deliver to a resistive load exactly one-half of the total power delivered to the same load by the input waveform.
- 4.30** Find the ratio between the average and rms value of the waveform shown in Figure P4.30.

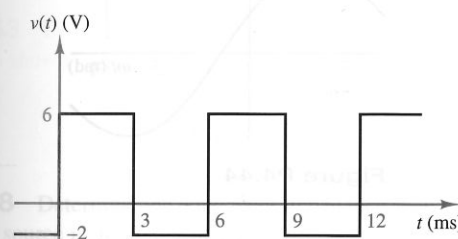


Figure P4.30

- 4.31** The current through a 1- $\Omega$  resistor is shown in Figure P4.31. Find the power dissipated by the resistor.

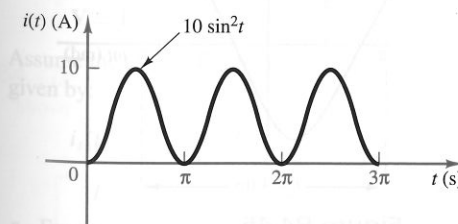


Figure P4.31

- 4.32** Derive the ratio between the average and rms value of the voltage waveform of Figure P4.32.

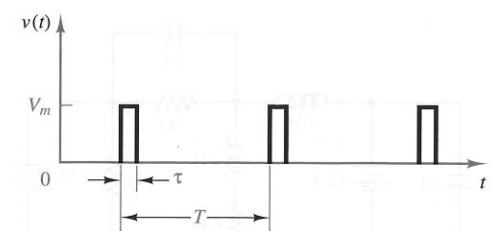


Figure P4.32

- 4.33** Find the rms value of the current waveform shown in Figure P4.33.

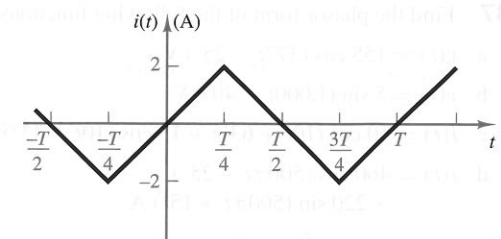


Figure P4.33

- 4.34** Determine the rms (or effective) value of  $v(t) = V_{DC} + v_{ac} = 35 + 63 \sin(215t^2)$  V

## Section 4.3 Circuits Containing Energy Storage Elements

- 4.35** Assume steady-state conditions and find the energy stored in each capacitor and inductor shown in Figure P4.35.

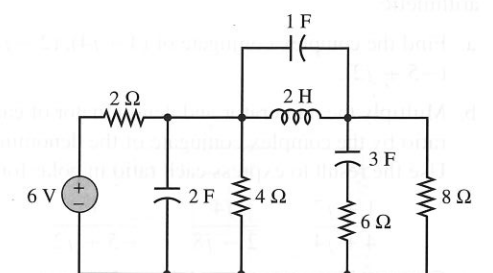


Figure P4.35

- 4.36** Assume steady-state conditions and find the energy stored in each capacitor and inductor shown in Figure P4.36.

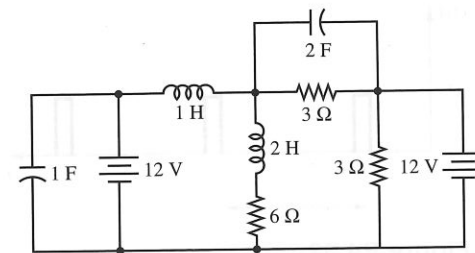


Figure P4.36

### Section 4.4: Phasor Solution of Circuits with Sinusoidal Sources

**4.37** Find the phasor form of the following functions:

- $v(t) = 155 \cos(377t - 25^\circ) \text{ V}$
- $v(t) = 5 \sin(1,000t - 40^\circ) \text{ V}$
- $i(t) = 10 \cos(10t + 63^\circ) + 15 \cos(10t - 42^\circ) \text{ A}$
- $i(t) = 460 \cos(500\pi t - 25^\circ) - 220 \sin(500\pi t + 15^\circ) \text{ A}$

**4.38** Convert the following complex numbers to polar form:

- $7 + j9$
- $-2 + j7$
- $j\frac{2}{3} + 4 - j\frac{1}{3} + 3$

**4.39** Convert the rectangular factors to polar form and compute the product. Also compute the product directly using the rectangular factors. Compare the results.

- $(50 + j10)(4 + j8)$
- $(j2 - 2)(4 + j5)(2 + j2)$

**4.40** Complete the following exercises in complex arithmetic.

- Find the complex conjugate of  $(4 + j4)$ ,  $(2 - j8)$ ,  $(-5 + j2)$ .
- Multiply the numerator and denominator of each ratio by the complex conjugate of the denominator. Use the result to express each ratio in polar form.

$$\frac{1 + j7}{4 + j4} \quad \frac{j4}{2 - j8} \quad \frac{1}{-5 + j2}$$

- Convert the numerator and denominator of each ratio in part b to polar form. Use the result to express each ratio in polar form.

**4.41** Convert the following expressions to rectangular form:

$$j + j \quad e^{-j\pi} \quad e^{+j2\pi}$$

**4.42** Find  $v(t) = v_1(t) + v_2(t)$  where

$$v_1(t) = 10 \cos(\omega t + 30^\circ)$$

$$v_2(t) = 20 \cos(\omega t + 60^\circ)$$

using:

- Trigonometric identities.
- Phasors.

**4.43** The current through and the voltage across a circuit element are, respectively,

$$i(t) = 8 \cos(\omega t + \frac{\pi}{4}) \text{ A}$$

$$v(t) = 2 \cos(\omega t - \frac{\pi}{4}) \text{ V}$$

where  $\omega = 600 \text{ rad/s}$ . Determine:

- Whether the element is a resistor, capacitor, or inductor.
- The value of the element in ohms, farads, or henrys.

**4.44** Express the sinusoidal waveform shown in Figure P4.44 using time-dependent and phasor notation.

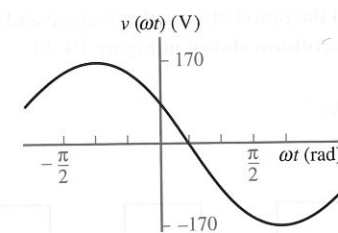


Figure P4.44

**4.45** Express the sinusoidal waveform shown in Figure P4.45 using time-dependent and phasor notation.

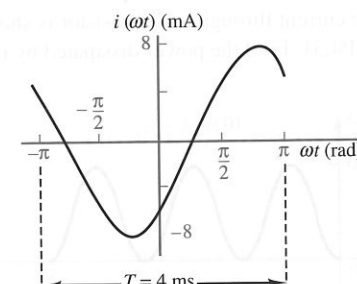


Figure P4.45

### Section 4.5 Impedance

**4.46** Convert the following pairs of voltage and current waveforms to phasor form. Each pair of waveforms

corresponds to an unknown element. Determine whether each element is a resistor, a capacitor, or an inductor, and compute the value of the corresponding parameter  $R$ ,  $C$ , or  $L$ .

- $v(t) = 20 \cos(400t + 1.2)$ ,  $i(t) = 4 \sin(400t + 1.2)$
- $v(t) = 9 \cos(900t - \frac{\pi}{3})$ ,  $i(t) = 4 \sin(900t + \frac{2}{3}\pi)$
- $v(t) = 13 \cos(250t + \frac{\pi}{3})$ ,  $i(t) = 7 \sin(250t + \frac{5}{6}\pi)$

**4.47** Determine the equivalent impedance seen by the source  $v_s$  in Figure P4.47 when:

$$v_s(t) = 10 \cos(4,000t + 60^\circ) \text{ V}$$

$$R_1 = 800 \text{ m}\Omega \quad R_2 = 500 \text{ n}\Omega$$

$$L = 200 \text{ mH} \quad C = 70 \text{ nF}$$

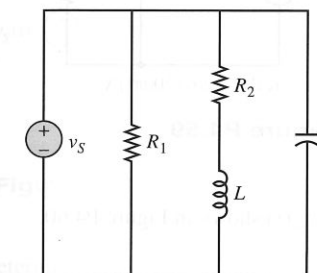


Figure P4.47

**4.48** Determine the equivalent impedance seen by the source  $v_s$  in Figure P4.47 when:

$$v_s(t) = 5 \cos(1,000t + 30^\circ) \text{ V}$$

$$R_1 = 300 \text{ m}\Omega \quad R_2 = 300 \text{ n}\Omega$$

$$L = 100 \text{ mH} \quad C = 50 \text{ nF}$$

**4.49** The generalized version of Ohm's law for impedance elements is:

$$\mathbf{V} = \mathbf{I} \mathbf{Z}$$

Assume the current through a  $0.5\text{-}\mu\text{F}$  capacitor is given by:

$$i_s(t) = I_o \cos(\omega t + \frac{\pi}{6})$$

$$I_o = 13 \text{ mA} \quad \omega = 1,000 \text{ rad/s}$$

- Express the source current in phasor notation.
- Determine the impedance of the capacitor.
- Determine the voltage across the capacitor, in phasor notation.

**4.50** Determine  $i_2(t)$  in the circuit shown in Figure P4.50. Assume:

$$i_1(t) = 100 \cos(\omega t + 4) \text{ mA}$$

$$i_3(t) = 80 \sin(\omega t - 1.2) \text{ mA}$$

$$i_4(t) = 150 \sin(\omega t + 2) \text{ mA}$$

$$\omega = 377 \text{ rad/s}$$

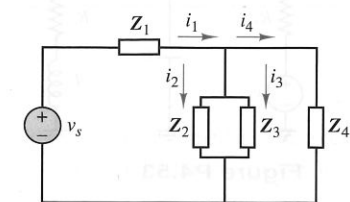


Figure P4.50

**4.51** Determine the voltage  $v_2(t)$  across  $R_2$  in the circuit of Figure P4.51.

$$i(t) = 20 \cos(533.33t) \text{ A}$$

$$R_1 = 8 \Omega \quad R_2 = 16 \Omega$$

$$L = 15 \text{ mH} \quad C = 117 \mu\text{F}$$

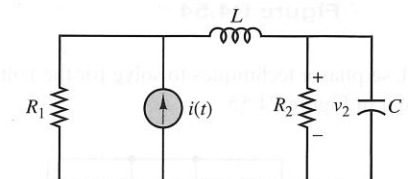


Figure P4.51

**4.52** Determine the frequency so that the current  $\mathbf{I}_i$  and the voltage  $\mathbf{V}_o$  in Figure P4.52 are in phase.

$$\mathbf{Z}_s = 13,000 + j\omega 3 \Omega$$

$$R = 120 \Omega$$

$$L = 19 \text{ mH} \quad C = 220 \text{ pF}$$

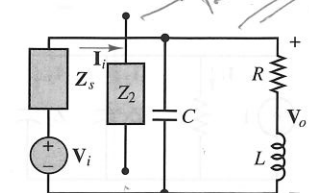


Figure P4.52

**4.53** A common model for a practical inductor is a coil resistance in series with an inductance  $L$ . The coil resistance accounts for the internal losses of an

inductor. Figure P4.53 shows an ideal capacitor in parallel with a practical inductor. Determine the current supplied by the source  $v_S$ . Assume:

$$v_S(t) = V_o \cos(\omega t + 0)$$

$$V_o = 10 \text{ V} \quad \omega = 6 \text{ Mrad/s} \quad R_S = 50 \Omega$$

$$R_C = 40 \Omega \quad L = 20 \mu\text{H} \quad C = 1.25 \text{ nF}$$

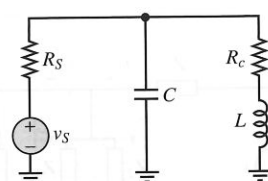


Figure P4.53

- 4.54** Use phasor techniques to solve for the current  $i(t)$  shown in Figure P4.54.

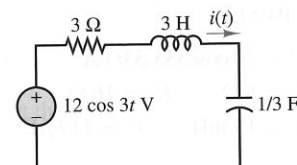


Figure P4.54

- 4.55** Use phasor techniques to solve for the voltage  $v(t)$  shown in Figure P4.55.

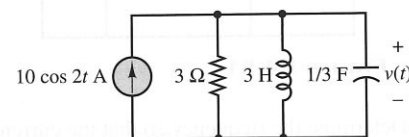


Figure P4.55

- 4.56** Solve for  $I_1$  in the circuit shown in Figure P4.56.

$$I = 20 \angle -\frac{\pi}{4} \text{ A} \quad R = 3 \Omega$$

$$Z_1 = -j3 \Omega \quad Z_2 = -j7 \Omega$$

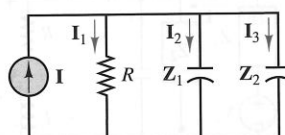


Figure P4.56

- 4.57** Solve for  $V_R$  shown in Figure P4.57. Assume:

$$\omega = 3 \text{ rad/s} \quad V_S = 13 \angle 0 \text{ V}$$

$$R = 15 \Omega \quad L_1 = 7 \text{ H} \quad L_2 = 2 \text{ H}$$

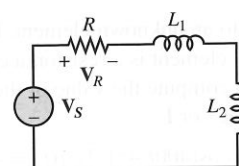


Figure P4.57

- 4.58** With reference to Problem 4.55, find the value of  $\omega$  for which the current through the resistor is maximum.

- 4.59** Find the current  $i_R(t)$  through the resistor shown in Figure P4.59.

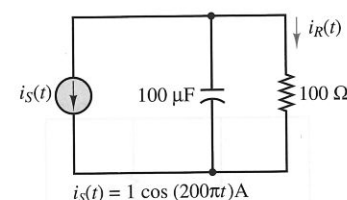


Figure P4.59

- 4.60** Find  $v_{out}(t)$  shown in Figure P4.60.

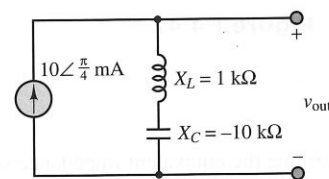


Figure P4.60

- 4.61** Find the impedance  $Z$  shown in Figure P4.61, assuming  $\omega = 2 \text{ rad/s}$ ,  $R_1 = R_2 = 2 \Omega$ ,  $C = 0.25 \text{ F}$ , and  $L = 1 \text{ H}$ .

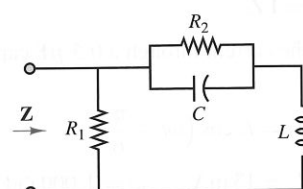
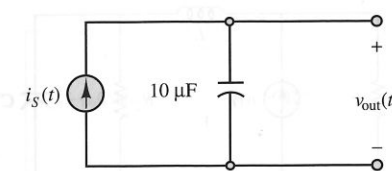
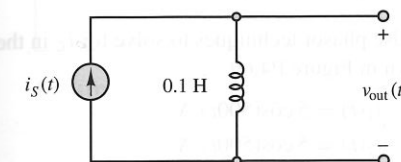


Figure P4.61

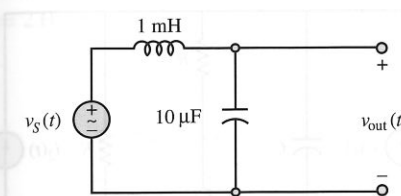
- 4.62** Find the sinusoidal steady-state output  $v_{out}(t)$  for each circuit shown in Figure P4.62.



$$(a) \quad i_S(t) = 10 \cos 100\pi t \text{ A}$$



$$(b) \quad i_S(t) = 20 \sin 10t \text{ A}$$



$$(c) \quad v_S(t) = 50 \sin 100t \text{ V}$$

Figure P4.62

- 4.63** Determine the voltage  $v_L(t)$  across the inductor shown in Figure P4.63.

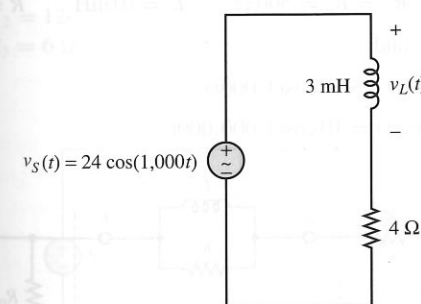


Figure P4.63

- 4.64** Determine the current  $i_R(t)$  through the resistor shown in Figure P4.64.

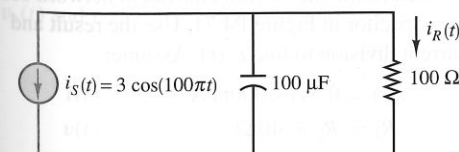


Figure P4.64

- 4.65** Find the frequency that causes the equivalent impedance  $Z_{eq}$  in Figure P4.65 to be purely resistive.

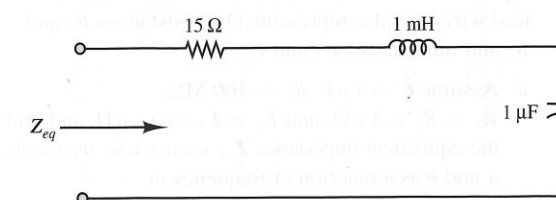
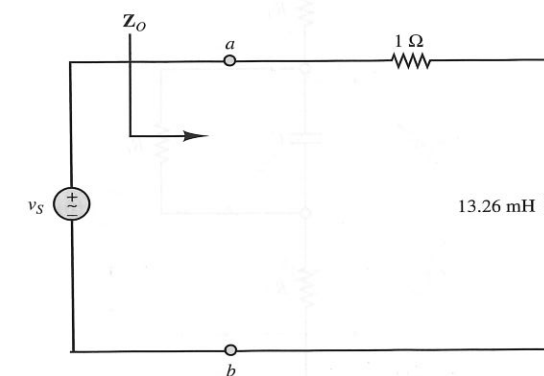


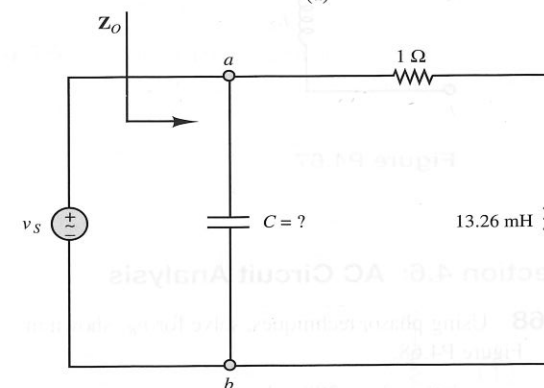
Figure P4.65

- 4.66**

- Find the equivalent impedance  $Z_o$  seen by the source in Figure P4.66(a). Assume the frequency is  $377 \text{ rad/s}$ .
- What capacitance should be placed between terminals  $a$  and  $b$ , as shown in Figure P4.66(b), to make the equivalent impedance  $Z_o$  purely resistive? [Hint: Find  $C$  so that the phase angle of  $Z_o$  is zero.]
- What is the amplitude of  $Z_o$  when the capacitor is included?



(a)



(b)

Figure P4.66

**4.67** A common model for a practical capacitor has a “leakage” resistance,  $R_C$ , in parallel with an ideal capacitor, as shown in Figure P4.67. The effects of lead wires are also represented by resistances  $R_1$  and  $R_2$  and inductances  $L_1$  and  $L_2$ .

- Assume  $C = 1 \mu\text{F}$ ,  $R_C = 100 \text{ M}\Omega$ ,  $R_1 = R_2 = 1 \mu\Omega$ , and  $L_1 = L_2 = 0.1 \mu\text{H}$ , and find the equivalent impedance  $Z_{ab}$  seen across terminals  $a$  and  $b$  as a function of frequency  $\omega$ .
- Find the range of frequencies for which  $Z_{ab}$  is capacitive.

[Hint: Assume that  $R_C$  is much greater than  $1/\omega C$  such that  $R_C$  can be ignored in part b.]

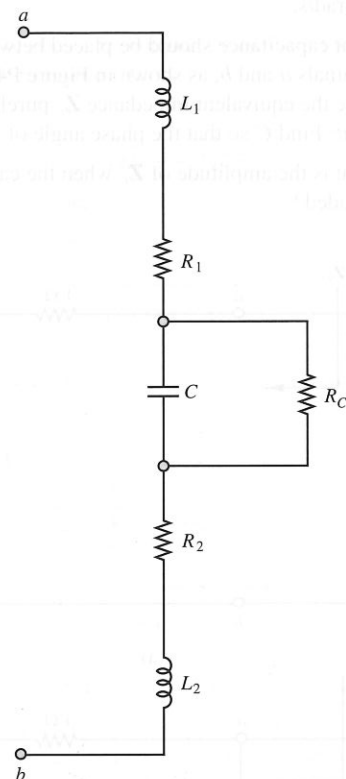


Figure P4.67

### Section 4.6: AC Circuit Analysis

**4.68** Using phasor techniques, solve for  $v_{R2}$  shown in Figure P4.68.

$$\begin{aligned} i(t) &= 3 \cos(200t) \text{ A} \\ R_1 &= 3 \Omega & R_2 &= 5 \Omega \\ L &= 18 \text{ mH} & C &= 170 \mu\text{F} \end{aligned}$$

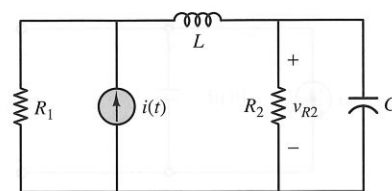


Figure P4.68

**4.69** Use phasor techniques to solve for  $i_L$  in the circuit shown in Figure P4.69.

$$\begin{aligned} i_1(t) &= 5 \cos(500t) \text{ A} \\ i_2(t) &= 5 \cos(500t) \text{ A} \\ R &= 5 \Omega & C &= 2 \text{ mF} & L &= 2 \text{ mH} \end{aligned}$$

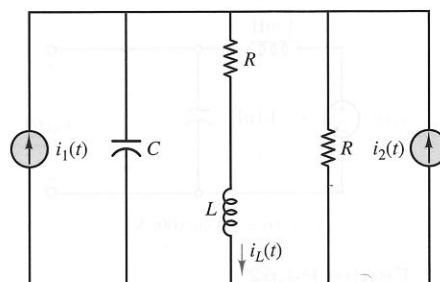


Figure P4.69

**4.70** Determine the Thévenin equivalent network seen by the load  $R_o$  in Figure P4.70. Assume:

$$R_S = R_o = 500 \Omega \quad L = 10 \text{ mH} \quad R = 1 \text{ k}\Omega$$

and:

- $v_S(t) = 10 \cos(1,000t)$
- $v_S(t) = 10 \cos(1,000,000t)$

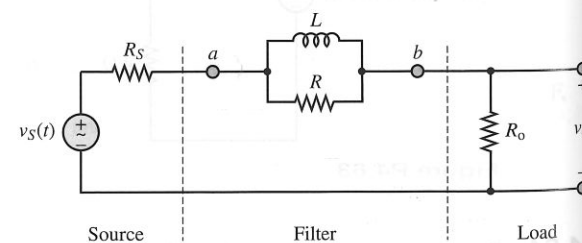


Figure P4.70

**4.71** Determine the Norton equivalent network seen by the capacitor in Figure P4.71. Use the result and current division to find  $i_C(t)$ . Assume:

$$\begin{aligned} i(t) &= 0.5 \cos(300t) \text{ A} \\ R_1 &= R_2 = 40 \Omega \\ L_1 &= L_2 = 200 \text{ mH} \\ C &= 15 \mu\text{F} \end{aligned}$$

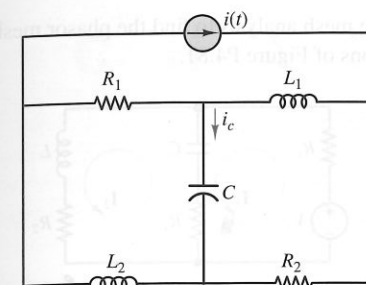


Figure P4.71

**4.72** Use phasor techniques to solve for  $i_L(t)$  in Figure P4.72. Assume  $v_S(t) = 2 \cos 2t$  V,  $R_1 = R_2 = 4 \Omega$ ,  $L = 2$  H, and  $C = 0.25$  F.

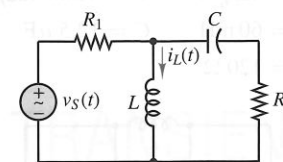


Figure P4.72

**4.73** Use mesh analysis to determine the currents  $i_1(t)$  and  $i_2(t)$  in Figure P4.73. Assume  $V_1 = 10e^{-j40}$  V,  $V_2 = 12e^{j40}$  V,  $R_1 = 8 \Omega$ ,  $R_2 = 4 \Omega$ ,  $R_3 = 6 \Omega$ ,  $X_L = 10 \Omega$ ,  $X_C = -14 \Omega$ .

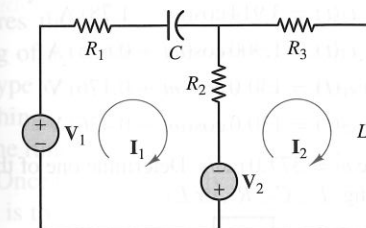


Figure P4.73

**4.74** Use nodal analysis to determine the node voltages  $v_a(t)$  and  $v_b(t)$  shown in Figure P4.74. Assume:

$$\begin{aligned} i(t) &= 2 \cos(300t) \text{ A} \\ v(t) &= 7 \cos(300t + \pi/4) \text{ V} \\ R_1 &= 4 \Omega & R_2 &= 3 \Omega & R_3 &= 5 \Omega \\ L &= 300 \text{ mH} & C &= 300 \mu\text{F} \end{aligned}$$

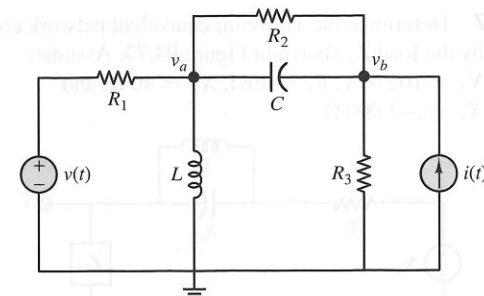


Figure P4.74

**4.75** The circuit shown in Figure P4.75 is a Wheatstone bridge, which can be used to determine the reactance  $X_4$  of an inductor or capacitor.  $R_1$  and  $R_2$  are adjusted until  $v_{ab}$  is zero.

- Assume a balanced bridge ( $v_{ab} = 0$ ) and determine  $X_4$  in terms of the other circuit elements.
- Assume a balanced bridge and let  $C_3 = 4.7 \mu\text{F}$ ,  $L_3 = 0.098$  H,  $R_1 = 100 \Omega$ ,  $R_2 = 1 \Omega$ , and  $v_S(t) = 24 \sin(2,000t)$ . What is the reactance of the unknown circuit element? Is it a capacitor or an inductor? What is its value?
- What frequency should be avoided by the source in this circuit, and why?

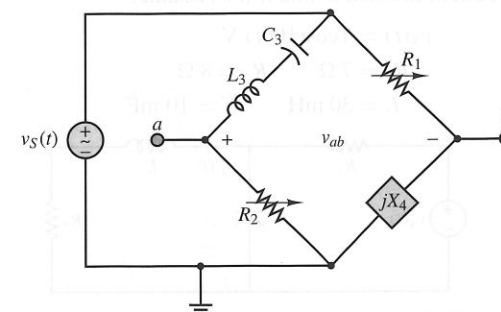


Figure P4.75

**4.76** Find the Thévenin equivalent network seen by the capacitor  $C$  in Figure P4.76. Use the result and voltage division to determine  $v_C(t)$ . Assume:

$$\begin{aligned} v(t) &= \cos(300t) \text{ V} \\ i(t) &= 2 \cos(300t) \text{ A} \\ R_1 &= 8 \Omega & R_2 &= 8 \Omega \\ L &= 3 \mu\text{H} & C &= 5 \mu\text{F} \end{aligned}$$

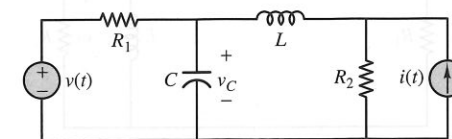


Figure P4.76